

EXPLAINABLE AI FOR BREAST CANCER DIAGNOSIS

ASSIGNMENT 2

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[Google Colab Link :](#)

Abstract

This project developed an explainable machine-learning prototype for breast cancer diagnosis using the Breast Cancer Wisconsin (Diagnostic) dataset. Random Forest, XGBoost, and Logistic Regression models were evaluated, with SHAP providing interpretable explanations. Results showed strong performance across all models, with XGBoost achieving the best results. Findings suggest that explainable AI can support more transparent breast cancer decision-making.

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1. Introduction

1.1 Background

Breast cancer is one of the most frequently diagnosed malignancies among women worldwide and remains a leading cause of cancer-related mortality despite advances in screening and treatment (Global Breast Cancer Statistics, 2024). Early detection and accurate characterisation of breast lesions are critical for selecting appropriate management strategies and improving long-term outcomes (Global Breast Cancer Statistics, 2024). Diagnostic pathways typically combine clinical examination, imaging modalities such as mammography and ultrasound, and tissue or cytological sampling; among these, fine-needle aspirate (FNA) cytology offers a minimally invasive method for obtaining cellular material from breast masses for microscopic assessment (Breast Cancer Cytology Review, 2017).

With the increasing digitisation of pathology workflows, FNA slides can be scanned and processed to extract numerical descriptors of cell nuclei morphology and texture. These quantitative features, which capture properties such as radius, perimeter, area, concavity and symmetry, provide a structured representation of cytological information that can be exploited by machine-learning algorithms (UCI Machine Learning Repository, 2021; Kaggle, 2016; GeeksforGeeks, 2024). The Breast Cancer Wisconsin (Diagnostic) (WDBC) dataset is a widely used benchmark that embodies this paradigm by providing derived features from digitised FNA images alongside benign or malignant labels for each case (UCI Machine Learning Repository, 2021; Kaggle, 2016).

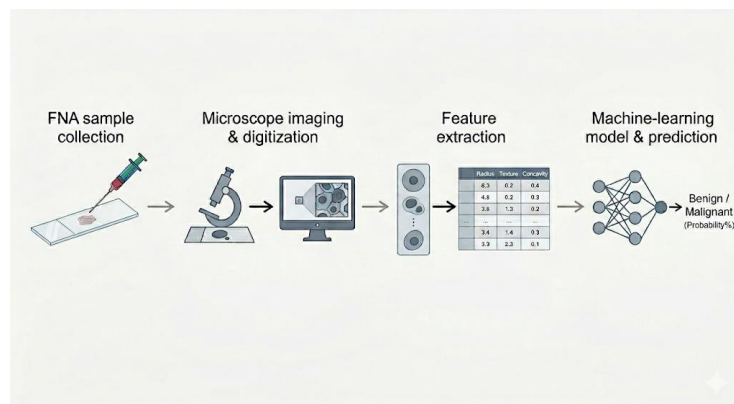


Fig.1. Diagnostic workflow from FNA acquisition to digitization, feature extraction and machine-learning-based prediction.

1.2 Problem context

A substantial body of research has demonstrated that machine-learning models can classify breast lesions using the WDBC dataset with high levels of accuracy, often exceeding 95% in controlled experiments (GeeksforGeeks, 2024; Predicting Breast Cancer Leveraging Supervised

ML, 2022; WDBC GFS Study, 2021). Algorithms such as logistic regression, support vector machines, decision trees, k-nearest neighbours, random forests and artificial neural networks have all been applied successfully to this problem, sometimes in combination with feature selection or ensemble techniques (Predicting Breast Cancer Leveraging Supervised ML, 2022; WDBC GFS Study, 2021). However, many of these efforts remain confined to offline scripts or notebooks, offering limited pathways for integration into clinical workflows.

A central barrier to adoption is the black-box nature of many high-performing models. Clinicians and other stakeholders often lack visibility into how models arrive at their predictions, which features drive malignancy decisions and whether outputs are robust in borderline or atypical cases (Explainable AI in Healthcare, 2025; Explainable AI for Healthcare Applications, 2023). This opacity can undermine trust and reduce the willingness of healthcare professionals to rely on automated decision-support tools, particularly in high-stakes contexts such as cancer diagnosis (Explainable AI in Healthcare, 2025). Moreover, many published works focus heavily on numerical performance metrics while giving less attention to user-centred interfaces that communicate predictions and their rationales in ways that align with clinical reasoning (Explainable AI in Healthcare, 2025; Supporting Breast Cancer Decisions, 2017).

1.3 Research aim and objectives

Explainable artificial intelligence (XAI) techniques, and SHapley Additive exPlanations (SHAP) in particular, have gained prominence as tools to make complex models more transparent by decomposing predictions into additive feature contributions (Explainable AI for Healthcare Applications, 2023; Explainable AI in Healthcare, 2025). When integrated thoughtfully into a user interface, such explanations have the potential to support clinicians in understanding, trusting and critically appraising automated recommendations (Explainable AI in Healthcare, 2025).

The overall **aim** of this project is therefore:

To develop and evaluate an explainable machine-learning–based decision-support prototype for breast cancer diagnosis using FNA-derived features, integrating SHAP-based explanations within a clinician-oriented web interface (adapted from UCI Machine Learning Repository, 2021 and Explainable AI in Healthcare, 2025).

To achieve this aim, the project pursues the following **objectives**:

- To preprocess and explore the Breast Cancer Wisconsin (Diagnostic) dataset in preparation for model training and evaluation (UCI Machine Learning Repository, 2021; Kaggle, 2016).
- To train and evaluate one or more supervised learning models for benign versus malignant classification using appropriate performance metrics (Predicting Breast Cancer Leveraging Supervised ML, 2022).

- To apply SHAP to generate both global feature importance and local, case-level explanations of model predictions (Explainable AI for Healthcare Applications, 2023; Explainable AI in Healthcare, 2025).
- To design and implement a web-based prototype that exposes predictions and SHAP-based explanations in a form accessible to clinicians and other end-users (Supporting Breast Cancer Decisions, 2017; Prototypical Decision-Supporting System, 2020).
- To analyse the predictive performance and interpretability of the system, discuss its limitations and identify opportunities for future work (Explainable AI in Healthcare, 2025).

1.4 Scope and structure of the report

The scope of this project is intentionally constrained to retrospective analysis of a single benchmark dataset and the development of a prototype decision-support interface. The models are trained exclusively on tabular features derived from digitised FNA images and do not incorporate raw imaging data, additional clinical risk factors or longitudinal patient information (UCI Machine Learning Repository, 2021; Kaggle, 2016). The prototype is not integrated with any hospital information system and is not intended for real clinical use; rather, it serves as a proof-of-concept artefact for exploring explainable machine learning in this context (Design Science Methodology, 2025; Design-Science Research Overview, 2025).

The remainder of this report is structured as follows. Chapter 2 reviews relevant literature on breast cancer diagnosis using the Breast Cancer Wisconsin datasets, machine-learning methods for this problem, explainable AI in healthcare and decision-support interfaces. Chapter 3 outlines the research methodology, including data description, preprocessing, model development and the SHAP-based explainability approach. Chapters 4 and 5 present the system design, implementation and experimental results. Chapter 6 discusses these findings in light of existing work, and Chapter 7 concludes the report and suggests directions for future research and development.

2. Literature Review

2.1 Breast cancer diagnosis and FNA cytology

Standard breast cancer diagnostic pathways combine clinical examination with imaging modalities such as mammography and ultrasound, often followed by tissue sampling for histopathological or cytological analysis when suspicious lesions are identified (Breast Cancer Cytology Review, 2017). FNA cytology plays an important role in this process by enabling minimally invasive sampling of breast masses via a fine needle, yielding cellular material for microscopic examination (Breast Cancer Cytology Review, 2017). Compared with core needle biopsy, FNA can be faster and less resource-intensive, although it may provide less architectural context (Breast Cancer Cytology Review, 2017).

Digital pathology has increasingly enabled the capture and storage of high-resolution images of FNA smears, which in turn facilitates quantitative image analysis. By segmenting cell nuclei and computing numerical descriptors of their shape, texture and distribution, it is possible to translate cytological observations into tabular data suitable for algorithmic analysis (UCI Machine Learning Repository, 2021; Kaggle, 2016; GeeksforGeeks, 2024). These developments have opened the door to computer-aided diagnostic systems that complement human expertise by identifying patterns in data that might be difficult to discern visually or consistently (WDBC GFS Study, 2021).

2.2 The Breast Cancer Wisconsin (Diagnostic) dataset

The Breast Cancer Wisconsin (Diagnostic) dataset (WDBC) is a widely used resource for evaluating classification algorithms in the context of breast cancer diagnosis (UCI Machine Learning Repository, 2021; Kaggle, 2016). The dataset was created at the University of Wisconsin and consists of features computed from digitised images of FNA of breast masses (UCI Machine Learning Repository, 2021). For each cell nucleus, ten real-valued features are calculated, including radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry and fractal dimension (UCI Machine Learning Repository, 2021; GeeksforGeeks, 2024). For each of these ten base features, the dataset provides the mean, standard error and “worst” (largest) values across nuclei in the image, resulting in a total of 30 features per case (UCI Machine Learning Repository, 2021; Kaggle, 2016).

The WDBC dataset contains 569 instances, of which 357 are labelled benign and 212 malignant (UCI Machine Learning Repository, 2021; Predicting Breast Cancer Leveraging Supervised ML, 2022). The class distribution is moderately imbalanced but still permits straightforward training of binary classifiers, especially when stratified sampling is used (Predicting Breast Cancer Leveraging Supervised ML, 2022). The dataset has become a standard benchmark in the machine-learning community due to its manageable size, well-documented feature set and relevance to a clinically important problem (UCI Machine Learning Repository, 2021; GeeksforGeeks, 2024). Numerous publicly accessible implementations and tutorials based on this dataset are available in repositories and educational platforms, further encouraging its reuse (GeeksforGeeks, 2024; Deepnote Notebook, 2023; WDBC GFS Study, 2021).

Item	Description
Dataset name	Breast Cancer Wisconsin (Diagnostic) (WDBC) archive.
Data source	UCI Machine Learning Repository; Kaggle mirror archive.
Number of instances (samples)	569 biopsy cases archive.
Class labels	Benign (B), Malignant (M) archive.
Number of benign cases	357 benign cases archive.
Number of malignant cases	212 malignant cases archive.

Class distribution (%)	≈62.8% benign, ≈37.2% malignant (based on 357 B, 212 M) archive.
Total number of predictive features	30 numeric features derived from FNA images archive.
Additional columns	1 ID column; 1 diagnosis column (B/M) archive.
Feature domains	Real-valued, strictly positive measurements
Base nuclear measurements (10)	Radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension archive.
Feature groups	For each of the 10 measurements, three statistics are provided: mean, standard error (SE), and “worst” (maximum) values, giving 30 total features archive.
Missing values	None reported in the standard WDBC dataset
Task type	Binary classification (predict whether the tumour is benign or malignant)

Table 1: Summary of the Breast Cancer Wisconsin (Diagnostic) (WDBC) dataset, including sample counts, class distribution and feature groups (mean, standard error and worst values for 10 nuclear measurements).

2.3 Machine learning for breast cancer diagnosis

A large body of work has investigated machine-learning models for classifying instances in the WDBC dataset as benign or malignant. Early studies applied methods such as linear discriminant analysis and k-nearest neighbours, achieving reasonable performance but leaving room for improvement (GeeksforGeeks, 2024). Subsequent research introduced more sophisticated models, including support vector machines with various kernels, decision trees, random forests and artificial neural networks (Predicting Breast Cancer Leveraging Supervised ML, 2022; WDBC GFS Study, 2021). Comparative analyses typically report high classification accuracies, often above 95%, with some combinations of feature selection and ensemble methods approaching near-perfect performance on held-out subsets (Predicting Breast Cancer Leveraging Supervised ML, 2022; Feature Selection Using Genetic Algorithm, 2016).

Feature selection has been a recurrent theme. For example, genetic algorithms and other search-based methods have been used to identify subsets of features that yield strong performance with simpler models, reducing dimensionality while preserving or even enhancing predictive power (Feature Selection Using Genetic Algorithm, 2016; WDBC GFS Study, 2021). These approaches often highlight that a smaller number of carefully chosen features can be sufficient to distinguish between benign and malignant cases, suggesting that not all 30 features are equally informative (Feature Selection Using Genetic Algorithm, 2016). Other works explore hybrid models, such as combining fuzzy logic with neural networks or integrating multiple classifiers through ensemble techniques (GeeksforGeeks, 2024; WDBC GFS Study, 2021).

Despite impressive metrics, many of these studies share similar limitations: they focus on offline performance on a single dataset, may not follow rigorous external validation protocols

and rarely address issues of interpretability or user-facing deployment (Explainable AI in Healthcare, 2025; Explainable AI for Healthcare Applications, 2023). As a result, there is a gap between research-grade models and tools that can realistically support and augment clinical decision-making in practice (Supporting Breast Cancer Decisions, 2017; Prototypical Decision-Supporting System, 2020).

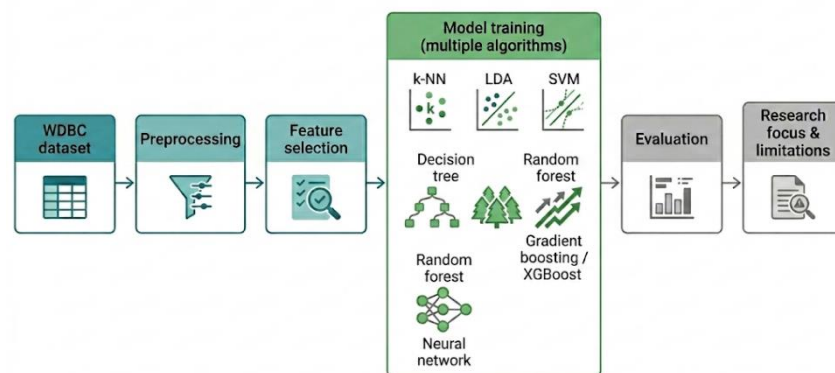


Fig. 2. Typical ML pipeline for the WDBC breast-cancer dataset.

2.4 Explainable AI in healthcare

Explainable artificial intelligence (XAI) has emerged as a critical area of research for domains where decisions have significant consequences and accountability requirements, such as healthcare, finance and law (Explainable AI in Healthcare, 2025; Explainable AI for Healthcare Applications, 2023). In medicine, clinicians and patients must be able to understand and interrogate algorithmic recommendations, assess whether they align with clinical knowledge and identify potential errors or biases (Explainable AI in Healthcare, 2025). A recent survey of XAI in healthcare highlights the growing use of methods such as SHAP, LIME and counterfactual explanations to interpret complex models across a range of conditions (Explainable AI in Healthcare, 2025).

SHAP provides a unified framework to explain individual predictions by attributing them to additive contributions from each feature, grounded in the concept of Shapley values from cooperative game theory (Explainable AI for Healthcare Applications, 2023). For each instance, SHAP computes how much each feature increases or decreases the predicted risk relative to a baseline, producing local explanations that are both model-agnostic in principle and consistent with several desirable properties (Explainable AI for Healthcare Applications, 2023). Aggregating SHAP values across many instances yields global explanations, such as feature importance rankings and summary plots, which reveal broader patterns in model behaviour (Explainable AI in Healthcare, 2025; Explainable AI for Healthcare Applications, 2023).

In healthcare applications, SHAP has been applied to problems such as predicting cardiovascular events, assessing diabetes complications and stratifying cancer risk (Explainable AI in Healthcare, 2025). Empirical studies suggest that SHAP-based explanations can enhance clinician trust and satisfaction, particularly when combined with clinically framed narrative text or structured explanation formats (Explainable AI in Healthcare, 2025). However, the interpretability of SHAP explanations depends strongly on how they are presented to users; raw plots or numerical values may require careful translation into domain-appropriate visualisations and narratives (Explainable AI in Healthcare, 2025).

2.5 Clinical decision-support tools and interfaces

Beyond algorithmic explainability, the literature emphasises the importance of user-centred design for clinical decision-support systems. Tools must integrate into workflows, use familiar terminology and present information at an appropriate level of detail for different user groups (Supporting Breast Cancer Decisions, 2017; Prototypical Decision-Supporting System, 2020). Interfaces that overload users with technical plots or metrics can be counterproductive, even if the underlying model is transparent from a technical perspective (Explainable AI in Healthcare, 2025). Conversely, carefully designed dashboards that summarise key information, visualise risk in intuitive forms and allow users to drill down into details where necessary are more likely to support effective decision-making (Supporting Breast Cancer Decisions, 2017).

Some works draw inspiration from established reporting standards such as BI-RADS in breast imaging, which codify how radiologists communicate risk and recommended actions (Breast Imaging Reporting Guidelines, 2019). Although the present project does not attempt to replicate BI-RADS or assign BI-RADS categories, the underlying principle of structured, graded risk communication is relevant for designing interfaces that present model outputs and explanations in a clinically meaningful way (Explainable AI in Healthcare, 2025). Existing XAI systems in healthcare still often prioritise algorithmic novelty over interface design, leaving substantial scope for research into explanation-aware user interfaces (Explainable AI in Healthcare, 2025; Prototypical Decision-Supporting System, 2020).

2.6 Identified gap and summary

The literature indicates that machine-learning models can achieve high accuracy on the Breast Cancer Wisconsin (Diagnostic) dataset and that methods such as SHAP offer powerful mechanisms for explaining predictions at both local and global levels (UCI Machine Learning Repository, 2021; GeeksforGeeks, 2024; Explainable AI in Healthcare, 2025). However, many prior studies treat explainability and user-facing design as secondary concerns, focusing chiefly on performance metrics or isolated visualisations rather than integrated, task-oriented tools (Explainable AI in Healthcare, 2025; Prototypical Decision-Supporting System, 2020). In particular, there is a shortage of prototype systems that combine a well-validated classifier, systematic SHAP-based explanations and an interface that presents these elements in a way

tailored to clinicians assessing FNA-derived breast lesion data (Supporting Breast Cancer Decisions, 2017; Prototypical Decision-Supporting System, 2020).

This project addresses this gap by developing an explainable decision-support prototype that builds on established modelling techniques for the WDBC dataset while foregrounding explanation and interface design. The subsequent chapters describe the methodology, implementation and evaluation of this artefact.

3. Research Methodology

3.1 Research approach

Given the dual focus on creating and evaluating a software artefact, this project adopts a design-science research (DSR) approach. Design-science is an artefact-oriented research paradigm developed primarily in information systems and software engineering that focuses on the rigorous creation and evaluation of innovative solutions to real-world problems (Design Science Methodology, 2025; Design-Science Research Overview, 2025). In the context of this work, the artefact is an explainable decision-support prototype for breast cancer diagnosis based on FNA-derived features.

The DSR process typically involves problem identification, definition of solution objectives, design and development of the artefact, demonstration in a relevant setting and evaluation, followed by communication of the results (Design-Science Research Overview, 2025). This project roughly follows these stages: the problem and objectives are established through the introduction and literature review; the artefact is designed and implemented as described in Chapter 4; and its behaviour is evaluated through quantitative model metrics and qualitative analysis of explanations and interface properties in Chapters 5 and 6 (Design Science Methodology, 2025).

The development process followed an iterative cycle in which model training and evaluation in a Google Colab environment preceded integration into the FastAPI backend and Next.js frontend, with repeated rounds of code refinement and interface adjustments based on inspection of predictions, explanations and error handling.

3.2 Functional and non-functional requirements

- Functional: prediction, SHAP explanations, feature-importance tab, model confidence tab, export report, login.
- Non-functional: security (MongoDB Atlas config), local deployment, acceptable latency, interpretability.

3.3 Data collection and description

The study uses the Breast Cancer Wisconsin (Diagnostic) dataset, obtained from a publicly available machine-learning repository such as the UCI Machine Learning Repository or Kaggle (UCI Machine Learning Repository, 2021; Kaggle, 2016). As noted earlier, the dataset contains 569 instances, each representing a breast mass assessed through FNA cytology, with 357 benign and 212 malignant cases (UCI Machine Learning Repository, 2021; Predicting Breast Cancer Leveraging Supervised ML, 2022). For each instance, 30 real-valued features are provided, derived from the ten underlying cell-nuclei descriptors captured as mean, standard error and “worst” values (UCI Machine Learning Repository, 2021; GeeksforGeeks, 2024).

Using a standard, well-documented dataset offers several advantages. It facilitates reproducibility, allows comparison with published studies using the same data and avoids ethical and legal complexities associated with handling identifiable patient records (UCI Machine Learning Repository, 2021; Kaggle, 2016). At the same time, reliance on a benchmark dataset imposes limitations in terms of representativeness and external validity, which are acknowledged and discussed in later chapters (Explainable AI in Healthcare, 2025; Explainable AI for Healthcare Applications, 2023).

3.4 Data preprocessing

Data preprocessing is a critical step in preparing the dataset for model training and explainability analysis. The WDBC dataset does not contain missing values in its standard form; however, basic checks are performed to confirm the absence of missing or anomalous entries (UCI Machine Learning Repository, 2021; GeeksforGeeks, 2024). Descriptive statistics and visualisations, such as histograms and correlation heatmaps, can be used to understand feature distributions and relationships (WDBC GFS Study, 2021).

Because the features span different scales and units, feature scaling is applied prior to training models that are sensitive to feature magnitudes, such as logistic regression and support vector machines (Predicting Breast Cancer Leveraging Supervised ML, 2022; WDBC GFS Study, 2021). A common approach is to standardise each feature to zero mean and unit variance based on the training set, applying the same transformation to the test set (GeeksforGeeks, 2024). Categorical labels (benign and malignant) are encoded as binary values for use with most machine-learning algorithms (GeeksforGeeks, 2024).

To evaluate the models, the dataset is partitioned into training and test subsets using a stratified split to preserve the class distribution in both sets. Typical configurations allocate 70–80% of the data for training and the remainder for testing, although the exact split ratio can be chosen based on project requirements (GeeksforGeeks, 2024; Predicting Breast Cancer Leveraging Supervised ML, 2022). Optionally, k-fold cross-validation within the training set can be employed for hyperparameter tuning and more robust performance estimation (Predicting Breast Cancer Leveraging Supervised ML, 2022).

3.5 Model development

The project trains and compares several supervised learning algorithms for the binary classification task of distinguishing benign from malignant lesions. Consistent with prior studies, candidate models may include logistic regression, support vector machines with radial basis function kernels, random forests, gradient boosting machines and simple feed-forward neural networks (Predicting Breast Cancer Leveraging Supervised ML, 2022; WDBC GFS Study, 2021). Each model is configured with reasonable initial hyperparameters and, where feasible, tuned using cross-validation to balance complexity and generalisation (Predicting Breast Cancer Leveraging Supervised ML, 2022).

Initial model development and evaluation were conducted in a Google Colab environment, where Random Forest, XGBoost and Logistic Regression classifiers were trained on the WDBC dataset and compared using train/test accuracy and confusion-matrix analysis. The best-performing versions of these models were then serialised and exported from Colab into the FastAPI backend as joblib artefacts, ensuring that the behaviour observed in notebooks is preserved in the deployed service.

Model performance is assessed using multiple metrics to capture different aspects of classification quality. Accuracy provides an overall measure of correct predictions, while precision and recall (sensitivity) focus on positive predictions and the detection of malignant cases, respectively (Predicting Breast Cancer Leveraging Supervised ML, 2022). Specificity quantifies the ability to correctly identify benign cases, and the F1-score summarises the balance between precision and recall (WDBC GFS Study, 2021). The area under the receiver operating characteristic curve (AUC) is also reported as a threshold-independent measure of discrimination performance (Predicting Breast Cancer Leveraging Supervised ML, 2022; Feature Selection Using Genetic Algorithm, 2016). Confusion matrices are generated for key models to visualise the distribution of true positives, true negatives, false positives and false negatives (WDBC GFS Study, 2021).

After comparing models on validation metrics, three classifiers—Random Forest, XGBoost and Logistic Regression—were retained for deeper analysis and integrated into the prototype, allowing users to switch between them and compare both performance metrics and explanation behaviour (Explainable AI for Healthcare Applications, 2023; Explainable AI in Healthcare, 2025). For example, tree-based models such as random forests and gradient boosting often work efficiently with tree-specific SHAP implementations (Explainable AI for Healthcare Applications, 2023).

3.6 Explainability method (SHAP)

To provide interpretable explanations for model predictions, the project employs SHapley Additive exPlanations (SHAP). SHAP treats the prediction of a model for a given instance as a cooperative game in which features are players and computes Shapley values that represent each feature's contribution to the difference between the instance's prediction and a baseline (Explainable AI for Healthcare Applications, 2023). These contributions sum to the prediction,

enabling additive decompositions that are both locally accurate and consistent with several intuitive properties (Explainable AI for Healthcare Applications, 2023).

For tree-based models, a tree-specific SHAP explainer can be used, which exploits the structure of decision trees to compute Shapley values efficiently (Explainable AI for Healthcare Applications, 2023). For other model types, such as kernel-based methods or neural networks, model-agnostic SHAP explainers approximate contributions through sampling and local linear surrogates (Explainable AI in Healthcare, 2025). In all cases, a background dataset—typically a subset of training instances—is required to define the reference distribution against which contributions are calculated (Explainable AI for Healthcare Applications, 2023).

The project uses SHAP at two levels. At the global level, SHAP values are aggregated across many instances to estimate the overall importance of each feature and to produce summary plots showing how feature values relate to their contributions to the prediction (Explainable AI in Healthcare, 2025; Explainable AI for Healthcare Applications, 2023). At the local level, SHAP values are computed for individual instances to generate case-specific explanations that highlight which features pushed the model towards a benign or malignant prediction and by how much (Explainable AI in Healthcare, 2025). These explanations form the basis for the user-facing explanation components in the prototype described in Chapter 4.

3.7 System evaluation strategy

The artefact is evaluated along two dimensions. First, quantitative evaluation focuses on the predictive performance of three candidate models—Random Forest, XGBoost and Logistic Regression—on the held-out WDBC test set, using accuracy, sensitivity, specificity, AUC-ROC and Brier score, along with the false negative rate (FNR) computed from the confusion matrices (Predicting Breast Cancer Leveraging Supervised ML, 2022; Feature Selection Using Genetic Algorithm, 2016). Reporting both discrimination metrics such as AUC-ROC and calibration metrics such as the Brier score follows recommendations from recent work comparing machine-learning models for breast-cancer prognostic prediction.

Second, qualitative evaluation examines the interpretability and usability of the SHAP-based explanations and the web interface. While full user studies with clinicians are beyond the scope of this project, the analysis considers whether the explanations are coherent with established knowledge about FNA-derived features, whether the interface presents information in a clear and structured manner and whether the system supports exploration of individual cases (Supporting Breast Cancer Decisions, 2017; Explainable AI in Healthcare, 2025). Consistent with the design-science research process, the prototype is instantiated as a running Next.js–FastAPI application in a local development environment, and its behaviour is assessed through end-to-end trials that exercise the prediction and explanation flows (Design Science Methodology, 2025; Design-Science Research Overview, 2025). These considerations are discussed in detail in Chapters 5 and 6.

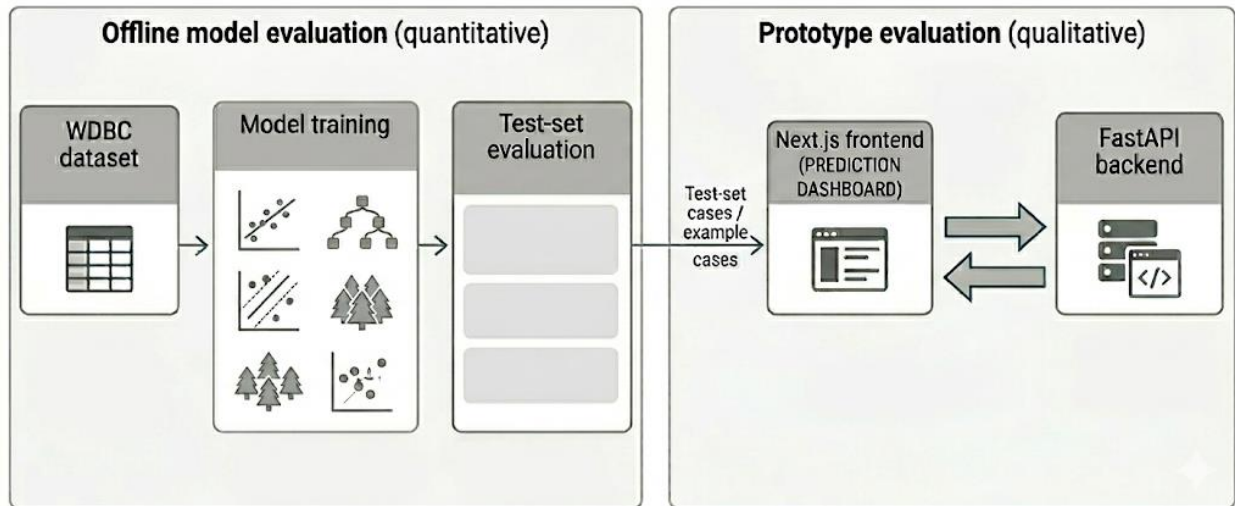


Figure .3. Evaluation setup overview – offline testing plus interface/explanation inspection using the Next.js + FastAPI prototype.

4. System Design and Implementation

4.1 Overall architecture

The decision-support prototype developed in this project is implemented as a web-based application that separates concerns across three main layers: a data and model layer, a FastAPI-based backend API and a Next.js frontend user interface. This architecture supports flexible deployment, enables reuse of the trained model in other contexts and allows the user interface to evolve independently of the underlying machine-learning code (Prototypical Decision-Supporting System, 2020; Design-Science Research Overview, 2025). The system operates on tabular feature vectors derived from FNA cytology, as defined by the WDBC dataset, and does not process raw images directly (UCI Machine Learning Repository, 2021; Kaggle, 2016).

At a high level, a typical interaction proceeds as follows. An authenticated user logs into the Next.js dashboard and either enters or reviews a set of WDBC-style feature values for a case using slider-based or numeric inputs. The frontend sends these values as a JSON payload to the FastAPI backend via a prediction endpoint. The backend applies the same preprocessing pipeline used during training, passes the data through the selected classifier and then computes SHAP values for the instance. The API responds with a structured JSON object containing the predicted class, probability and a vector of SHAP contributions for each feature, which the frontend renders in multiple explanation views (Explainable AI in Healthcare, 2025; Breast Cancer Prediction SHAP Dashboard, 2025).

4.2 Backend implementation

The backend is implemented as a REST-style web service using **FastAPI** with Python. The main application module imports typed utilities and core libraries including:

```
python
from typing import Dict, Any, List, Optional
import numpy as np
import joblib
from fastapi import FastAPI, HTTPException
from fastapi.middleware.cors import CORSMiddleware
from pydantic import BaseModel
from summary_generator import summary_generator, FeatureContribution
import shap
```

This structure ensures that request and response bodies are described by Pydantic models, which are automatically validated and documented by FastAPI. A serialised classifier—trained using scikit-learn, TensorFlow-CPU or xgboost—is stored on disk using joblib and loaded into memory on startup, together with the fitted preprocessing pipeline (for example, scalers and encoders). The project's Python environment is managed through a requirements.txt file that includes fastapi, uvicorn[standard], numpy, pandas, matplotlib, scikit-learn (version 1.6.1), tensorflow-cpu, shap, lime, joblib and xgboost, facilitating reproducible deployment.

To support explainability, the backend constructs a SHAP explainer instance appropriate to the chosen model. For tree-based ensemble models, a tree-specific SHAP explainer is used to compute feature attributions efficiently (Explainable AI for Healthcare Applications, 2023). For each prediction request, the backend computes local SHAP values for the supplied instance relative to a small background dataset derived from the training data. The API returns both the raw SHAP vector and a transformed, human-readable summary generated by the summary_generator component, which merges feature contributions into short textual explanations aligned with clinical concepts (Explainable AI in Healthcare, 2025).

The backend is containerised in a Docker image that bundles the Python runtime, dependencies and FastAPI application. A Uvicorn ASGI server runs the app, and CORS is configured via CORSMiddleware to allow cross-origin requests from the Next.js frontend origin while blocking others. Basic error handling ensures that invalid or incomplete payloads result in informative HTTP error responses rather than silent failures.

```
main.py 8 X
Explainable-AI-For-Breast-Cancer-Diagnosis > my-backend > main.py > get_global_importance

1 # main.py
2 from typing import Dict, Any, List, Optional
3
4 import numpy as np
5 import joblib
6 from fastapi import FastAPI, HTTPException
7 from fastapi.middleware.cors import CORSMiddleware
8 from pydantic import BaseModel
9 from summary_generator import summary_generator, FeatureContribution
10 import shap
11
12 app = FastAPI()
13
14 origins = ["http://localhost:3000", "http://127.0.0.1:3000"]
15 app.add_middleware(
16     CORSMiddleware,
17     allow_origins=origins,
18     allow_credentials=True,
19     allow_methods=["*"],
20     allow_headers=["*"],
21 )
22
23 # =====
24 # Load artifacts (TOP 10)
25 # =====
26
27 # RF model (main explainable model)
28 rf_model = joblib.load("bc_rf_model.pkl")
29
30 # XGBoost + Logistic Regression (additional comparators)
31 xgb_model = joblib.load("bc_xgb_model.pkl")
32 lr_model = joblib.load("bc_lr_model.pkl")
33
34 scaler = joblib.load("scaler.pkl")
35 feature_names = np.load("feature_names.npy", allow_pickle=True) # len = 10
36 background_scaled = np.load("background_scaled.npy", allow_pickle=True)
37 rf_global_importance = np.load("global_importance_rf.npy", allow_pickle=True).tolist()
38 xgb_global_importance = np.load("global_importance_xgb.npy", allow_pickle=True).tolist()
39 lr_global_importance = np.load("global_importance_lr.npy", allow_pickle=True).tolist()
40
41 feature_stats: Dict[str, Dict[str, float]] = np.load(
42     "feature_stats.npy", allow_pickle=True
43 ).item()
44
45 benign_baseline = np.load("benign_baseline.npy") # shape (10,)
46
47 feature_index = {name: i for i, name in enumerate(feature_names)}
48 IDX_RADIUS = feature_index["mean radius"]
49 IDX_TEXTURE = feature_index["mean texture"]
50 IDX_CONCAVITY = feature_index["mean concavity"]
51 IDX_MEAN_PERIM = feature_index["mean perimeter"]
52 IDX_MEAN_CP = feature_index["mean concave points"]
53 IDX_WORST_RADIUS = feature_index["worst radius"]
54 IDX_WORST_PERIM = feature_index["worst perimeter"]
55 IDX_WORST_AREA = feature_index["worst area"]
56 IDX_WORST_CP = feature_index["worst concave points"]
57 IDX_WORST_CONCAVITY = feature_index["worst concavity"]
58
```

```

59 required = ["mean radius", "mean texture", "mean concavity"]
60 missing = [f for f in required if f not in feature_index]
61 if missing:
62     raise ValueError(f"Missing required slider features in feature_names.npy: {missing}")
63
64 # SHAP explainers per model
65 rf_explainer = shap.TreeExplainer(rf_model, background_scaled)
66 xgb_explainer = shap.TreeExplainer(xgb_model, background_scaled)
67 # For LR, use LinearExplainer
68 lr_explainer = shap.LinearExplainer(lr_model, background_scaled)
69
70 # =====
71 # Helper functions
72 # =====
73
74 def predict_proba_single_rf(x_raw: np.ndarray) -> float:
75     x_scaled = scaler.transform([x_raw])
76     proba = rf_model.predict_proba(x_scaled)[0][1] # class 1 = benign
77     return float(proba)
78
79 def predict_proba_single_xgb(x_raw: np.ndarray) -> float:
80     x_scaled = scaler.transform([x_raw])
81     proba = xgb_model.predict_proba(x_scaled)[0][1] # benign
82     return float(proba)
83
84 def predict_proba_single_lr(x_raw: np.ndarray) -> float:
85     x_scaled = scaler.transform([x_raw])
86     proba = lr_model.predict_proba(x_scaled)[0][1] # benign
87     return float(proba)

```

```

88
89 def compute_shap_for_x(x_raw: np.ndarray, model_key: str = "RF") -> np.ndarray:
90     """
91     Compute SHAP values for a single sample using the chosen model.
92     model_key: "RF", "XGB", or "LR"
93     """
94     x_scaled = scaler.transform([x_raw])
95
96     model_key = model_key.upper()
97     if model_key == "RF":
98         shap_out = rf_explainer.shap_values(x_scaled)
99     elif model_key == "XGB":
100         shap_out = xgb_explainer.shap_values(x_scaled)
101     elif model_key == "LR":
102         shap_out = lr_explainer.shap_values(x_scaled)
103     else:
104         raise ValueError(f"Unknown model_key for SHAP: {model_key}")
105
106     # For tree models, shap_out may be a list [class0, class1]
107     if isinstance(shap_out, list):
108         # Take benign class = 1
109         shap_values = shap_out[1][0]
110     else:
111         # LinearExplainer typically returns a 2D array (n_samples, n_features)
112         shap_values = shap_out[0]
113
114     shap_values = np.array(shap_values, dtype=float)
115     if shap_values.ndim > 1:
116         shap_values = shap_values.reshape(-1)

```

```
main.py 8 X
Explainable-AI-For-Breast-Cancer-Diagnosis > my-backend > main.py > ...

318 # =====
319 # Schemas & endpoints
320 # =====
321
322 class PredictRequest(BaseModel):
323     radius: float
324     texture: float
325     concavity: float
326     mean_perimeter: float
327     mean_concave_points: float
328     worst_radius: float
329     worst_perimeter: float
330     worst_area: float
331     worst_concave_points: float
332     worst_concavity: float
333     explanationMode: Optional[str] = None
334     model: Optional[str] = "RF"
335
336 class PredictResponse(BaseModel):
337     prediction_label: str
338     malignant_probability: float
339     benign_probability: float
340     model: dict
341     mode2: dict
342     mode3: dict
343     model_comparisons: list
344
345 class GlobalFeature(BaseModel):
346     feature: str
347     importance: float
348     group: str
349     rank: int
350
351 @app.post("/predict", response_model=PredictResponse)
352 def predict(req: PredictRequest):
353     x_raw = benign_baseline.copy()
354     x_raw[IDX_RADIUS] = req.radius
355     x_raw[IDX_TEXTURE] = req.texture
356     x_raw[IDX_CONCAVITY] = req.concavity
357     x_raw[IDX_MEAN_PERIM] = req.mean_perimeter
358     x_raw[IDX_MEAN_CP] = req.mean_concave_points
359     x_raw[IDX_WORST_RADIUS] = req.worst_radius
360     x_raw[IDX_WORST_PERIM] = req.worst_perimeter
361     x_raw[IDX_WORST_AREA] = req.worst_area
362     x_raw[IDX_WORST_CP] = req.worst_concave_points
363     x_raw[IDX_WORST_CONCAVITY] = req.worst_concavity
364
365     model_key = (req.model or "RF").upper()
366     return build_payload_for_single(x_raw, model_key=model_key)
```

Fig.4. FastAPI prediction endpoint combining model inference and SHAP-based explanation modes.

```
prediction-dashboard.tsx 9+ X
Explainable-AI-For-Breast-Cancer-Diagnosis > my-frontend > components > ui > prediction-dashboard.tsx > PredictionDashboard

1 // components/PredictionDashboard.tsx
2 import type { PredictionResponse } from "@app/PredictionPage";
3 import Model from "../model";
4 import Mode2 from "../mode2";
5 import Mode3 from "../mode3";
6 import { useEffect, useState } from "react";
7
8 interface Props {
9     result: PredictionResponse;
10    explanationMode: string;
11    selectedModel: "RF" | "XGB" | "LR";
12 }
13
14 export default function PredictionDashboard({ result, explanationMode, selectedModel }: Props) {
15     const primaryModel = result.model_comparisons.find(
16         (m) => m.short_name === selectedModel
17     ) ?? {
18         malignant_probability: result.malignant_probability,
19         benign_probability: result.benign_probability,
20     };
21
22     const malignantPct = primaryModel.malignant_probability * 100;
23     const benignPct = primaryModel.benign_probability * 100;
```

Fig.5 : React prediction dashboard component that renders AI diagnosis, malignancy probability and multiple explanation modes.

4.3 Frontend and user interface

The frontend is implemented using Next.js with TypeScript, styled using CSS (for example CSS modules or a utility framework), and deployed as a single-page dashboard. The main interface is organised into three primary areas: a data input panel, a prediction summary and case-level explanation panel, and a set of additional tabs that expose global feature importance, model confidence and report export functionality (Breast Cancer Prediction SHAP Dashboard, 2025). This organisation supports clinical workflows where users first supply or verify data, then review a high-level risk estimate, and finally inspect feature-level and model-level details as needed.

The main prediction view organises the interface into three explanation-aware sections: a data input sidebar, a central prediction and risk overview, and a lower area that hosts multiple explanation modes. The input sidebar contains patient and case identifiers, slider and numeric controls for the key WDBC-derived features (for example, mean radius and worst concave points) and compact clinical reference ranges that indicate typical benign and malignant value intervals, helping users to judge whether a given measurement is low, typical or high in a cytological sense.

Within the prediction area, the dashboard offers three complementary explanation modes that present model outputs in different formats suited to clinical reasoning. Mode 1 (“Text Summary”) displays a text-first view in which the top SHAP-ranked features are rendered as coloured cards with plain-language descriptions using FNA Categories, risk-colour indicators and a generated narrative summary that synthesises the dominant pattern for the current case. Mode 2 (“Bars + Text”) focuses on a compact horizontal bar chart that ranks the most influential features by percentage impact and direction (towards malignancy or benignity), while Mode 3 (“Feature impact”) presents a longer ranked list of all analysed features, allowing users to see where specific measurements sit in the overall contribution profile.

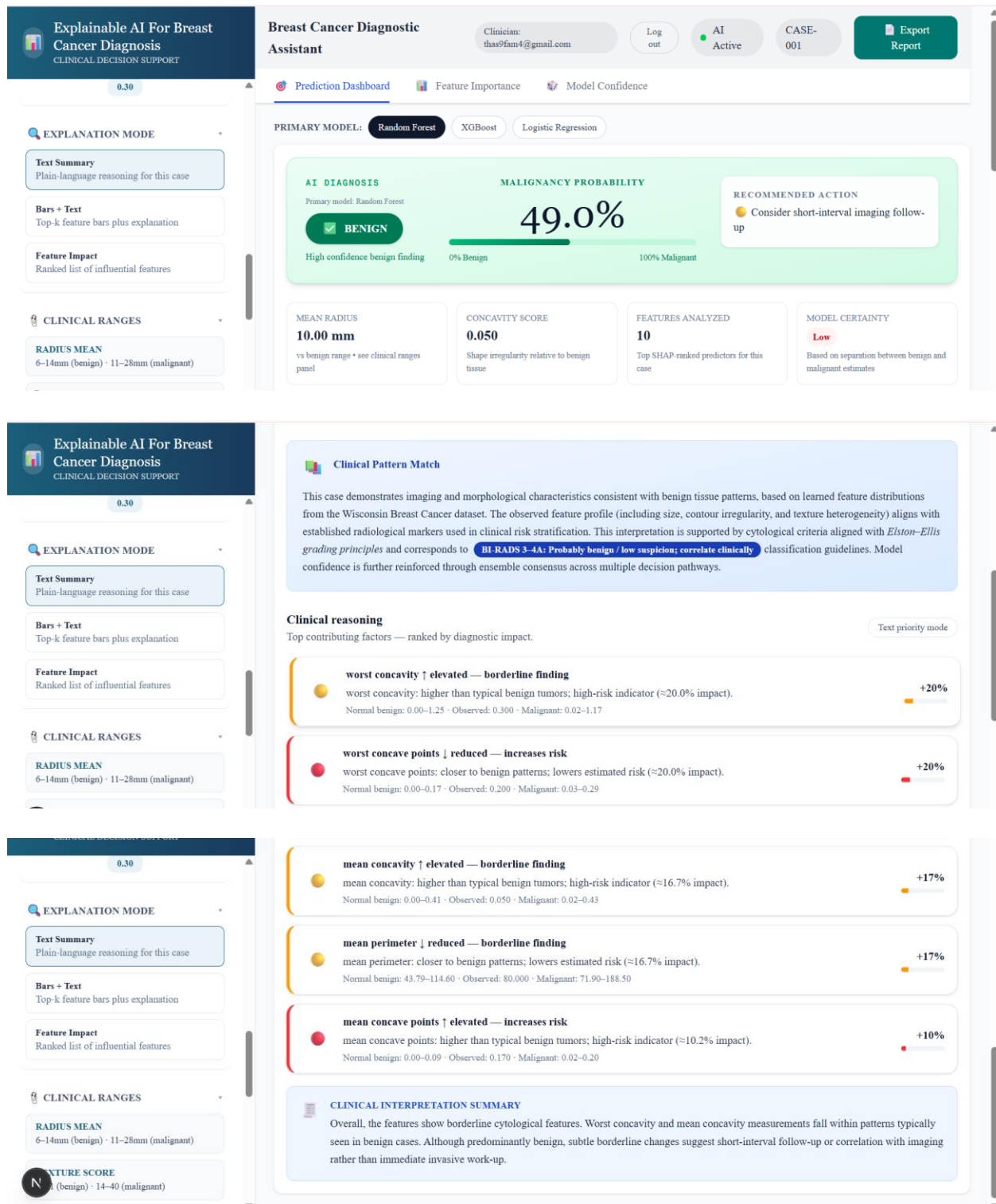


Fig.6. Explanation Mode 1 (“Clinical reasoning – text summary”)

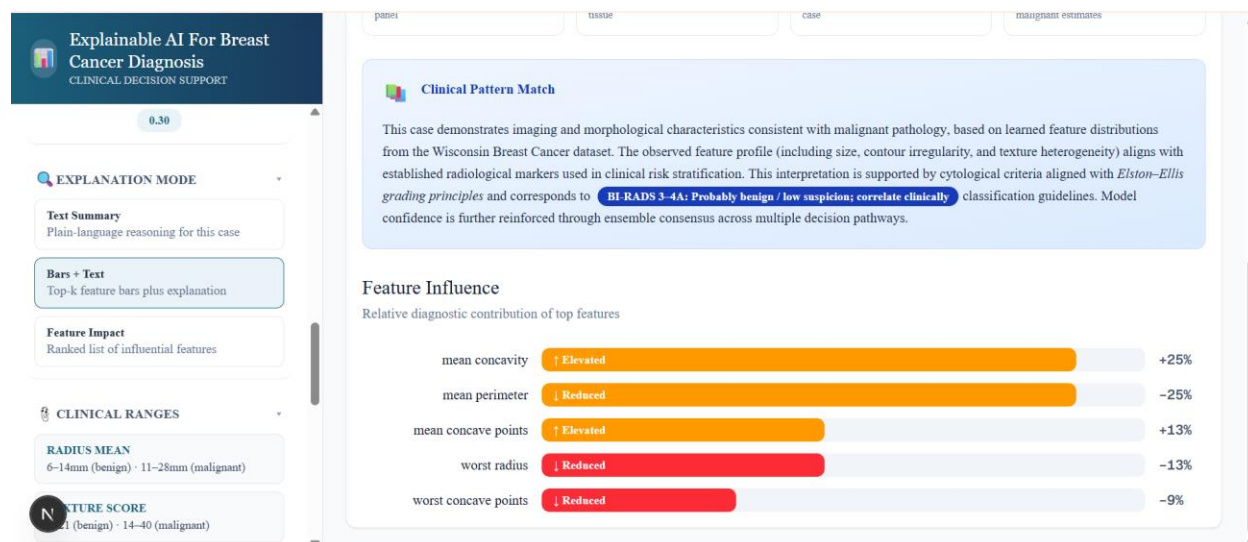


Fig.7. Explanation Mode 2 (“Feature influence – bars + text”)

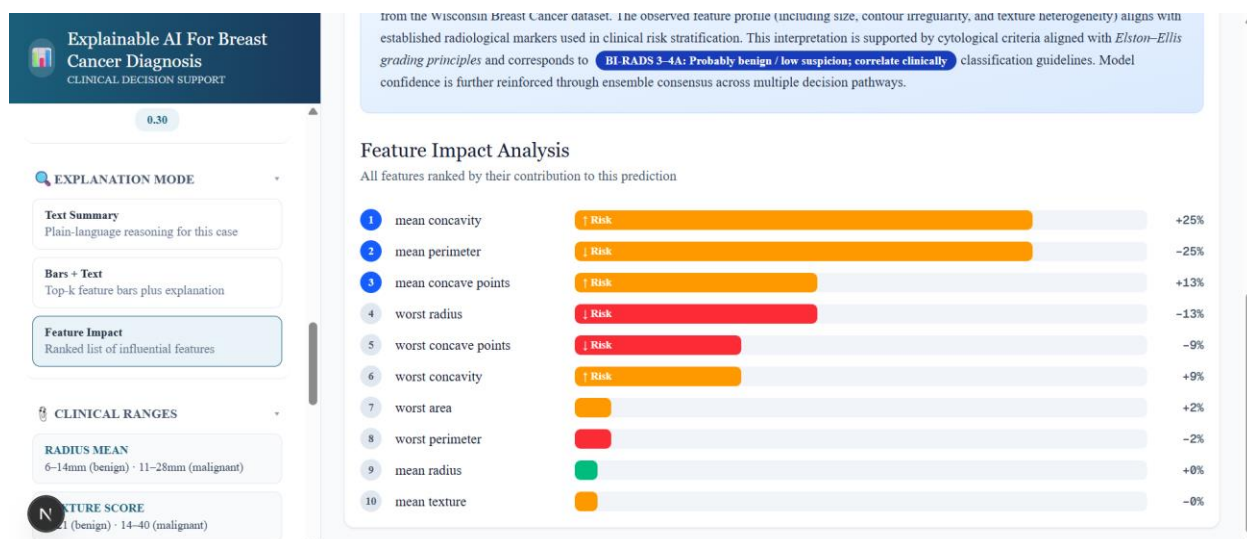


Fig.8. Explanation Mode 3 (“Feature impact analysis”)

Beneath the primary risk header, a dedicated “clinical pattern-match” panel summarises how the observed feature profile compares with benign versus malignant patterns learned from the Wisconsin dataset and expresses this in BI-RADS-inspired risk language, without claiming to assign formal BI-RADS scores.

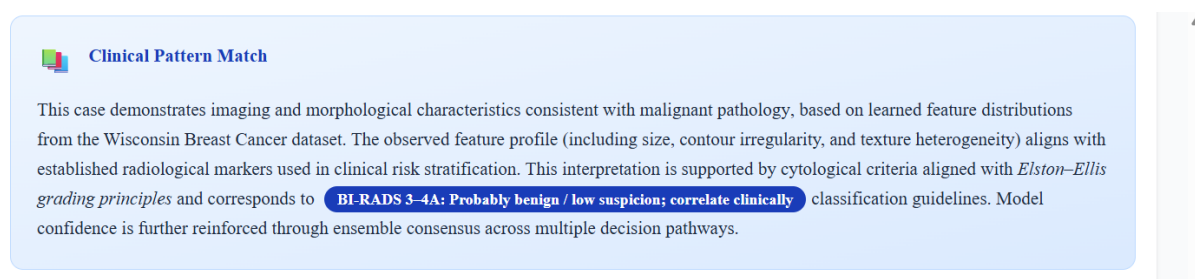


Fig.9. Clinical Pattern Match (Bi-Rads Risk Style Summary)

At the top of the page, a navigation bar provides access to **model toggles**, session controls and additional analysis views. Users can switch between the three trained models (Random Forest, XGBoost and Logistic Regression) using a simple toggle, allowing side-by-side comparison of predictions and explanations for the same case. The top bar also includes a logout button, an **Export report** action, AI status element that reflects the current primary model and its prediction state, and a **Feature importance** tab that opens a view where users can filter global SHAP plots by feature group (mean, worst, standard error or “all features”), thereby focusing on the most clinically relevant subset of the 30 WDBC features at a time.

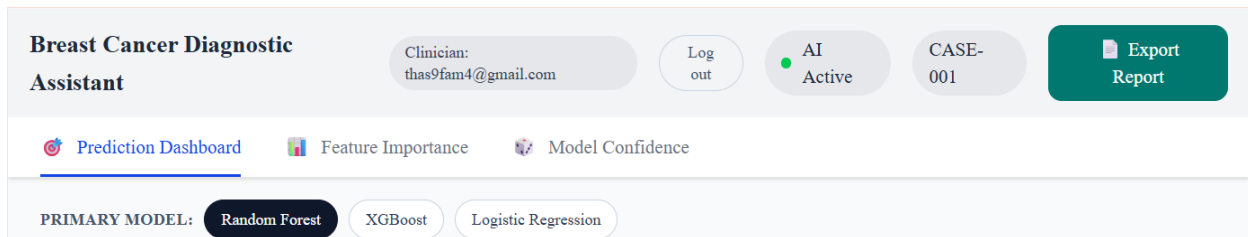


Fig.10. Top Nav

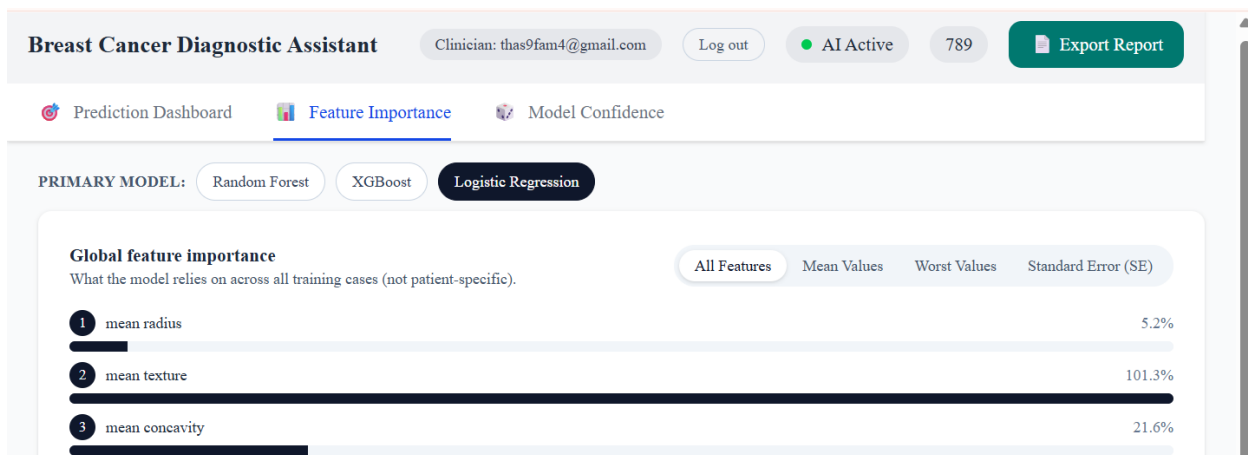


Fig.11. Feature Importance Tab

A persistent **sidebar** on the left groups together key controls and contextual information. It shows patient and case identifiers, the full list of WDBC-derived features with slider and numeric inputs, and concise **clinical reference ranges** for each feature to help users judge whether a given value is low, typical or high for benign and malignant FNA patterns. The sidebar also exposes basic model information (for example, which of the three models is currently selected) so that users can see at a glance which algorithm produced the active prediction and explanations. Each control is strongly typed in TypeScript and bound to a form model that mirrors the Pydantic schema used by the FastAPI backend, reducing the likelihood of mismatches between frontend and backend. On submission, the dashboard calls the FastAPI prediction endpoint and displays the resulting class (benign or malignant), malignancy and

benign probabilities and a short textual summary in the prediction panel, giving an immediate sense of model confidence for the current case.

Explainable AI For Breast Cancer Diagnosis
CLINICAL DECISION SUPPORT

PATIENT INPUT

Patient / Sample ID
CASE-001

Adjust sliders to match your patient; see Clinical ranges below for benign/malignant values.

MEAN RADIUS (mm)
6mm 30mm
10.0 mm

TEXTURE SCORE
9 40
15.0

CONCAVITY SCORE
0.50
0.05

N

Fig.12. Input and sliders

Explainable AI For Breast Cancer Diagnosis
CLINICAL DECISION SUPPORT

CLINICAL RANGES

RADIUS MEAN
6–14mm (benign) · 11–28mm (malignant)

TEXTURE SCORE
9–21 (benign) · 14–40 (malignant)

CONCAVITY SCORE
0–0.13 (benign) · 0.07–0.43 (malignant)

MEAN PERIMETER
40–100 (benign) · 70–200 (malignant)

MEAN CONCAVE POINTS
0.00–0.05 (benign) · 0.03–0.20 (malignant)

WORST RADIUS
10–20mm (benign) · 18–40mm (malignant)

WORST PERIMETER
150 (benign) · 120–300 (malignant)

N

Fig.13. Clinical ranges

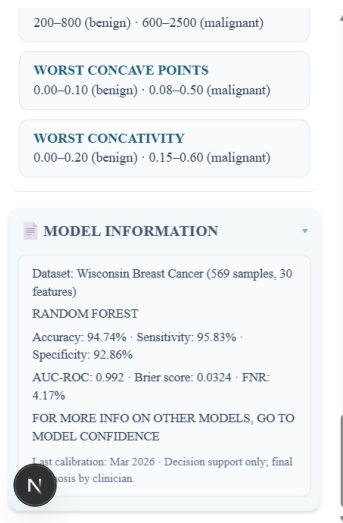


Fig.14. Model info

Below this summary, a **local explanation** view presents SHAP information for the selected case as horizontal bar charts or ordered lists, showing which features pushed the prediction towards malignancy or benignity and by how much. In addition, a dedicated **Feature importance** tab provides **global SHAP feature-importance plots** for the ten WDBC feature groups computed across all test-set cases and for each of the three evaluated models (Random Forest, XGBoost and Logistic Regression). This allows users to compare how strongly each model relies on specific morphological descriptors when distinguishing benign from malignant lesions, in line with recommendations to expose both global and local explanations.

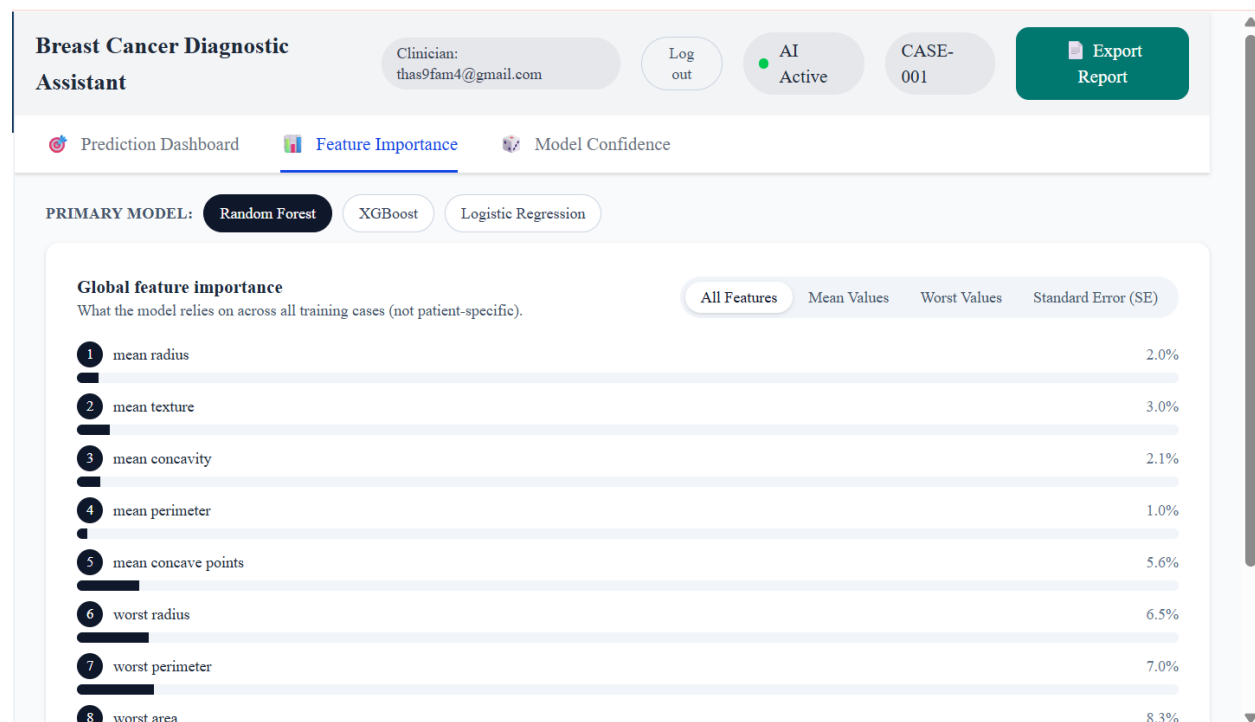


Fig.15. A Feature Importance Tab

A separate **Model confidence** tab aggregates information about model-level behaviour. This view shows test-set performance metrics per model (accuracy, sensitivity, specificity, AUC-ROC, Brier score and false negative rate), confusion matrices and indicators of ensemble agreement or disagreement between the three models for the current case. Presenting these elements together is intended to help users judge not only the predicted label, but also how certain the models are and how consistently they behave across different algorithms (The Application and Comparison of Machine Learning Models, 2022; Detection of Breast Cancer Using Machine Learning and Explainable AI, 2025).

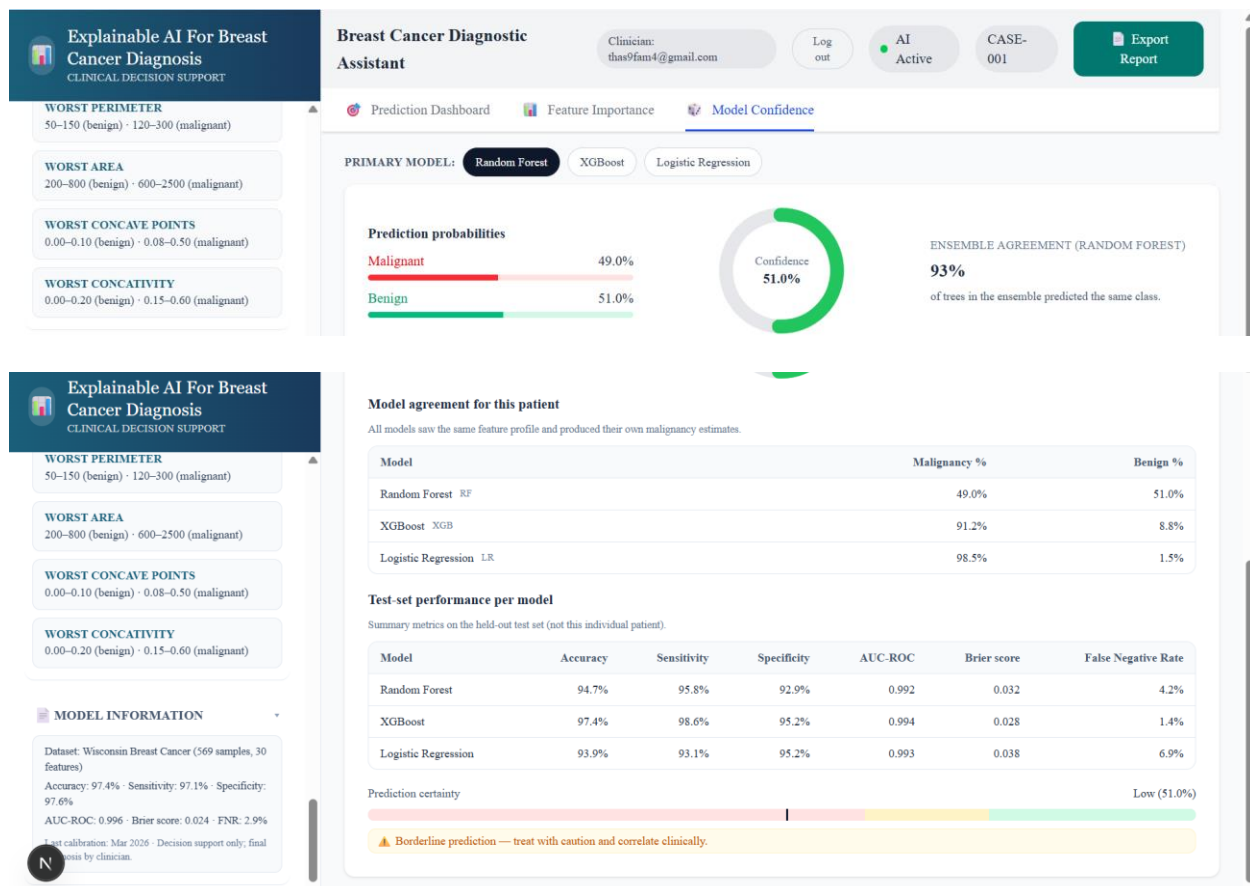


Fig.16. Model confidence Tab

Finally, the dashboard includes an **Export report** action that generates a structured diagnosis report for the current case, similar in content and layout to the example in CASE-001-report-7.pdf. The exported report summarises the primary model, predicted label, malignancy and benign probabilities and a narrative explanation generated by the summary component, along with a disclaimer that the output is for decision support only and not a substitute for professional clinical judgement. This feature illustrates how XAI outputs can be packaged into artefacts that might be incorporated into clinical documentation in future work.

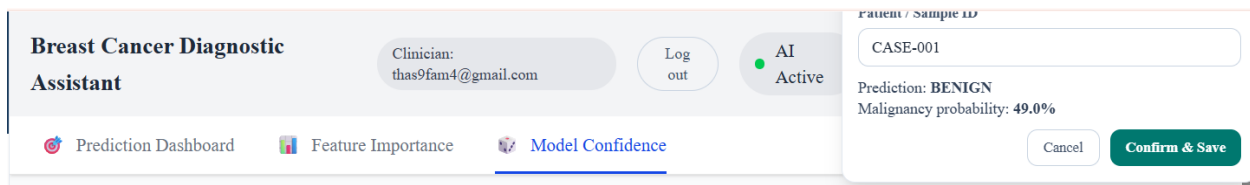


Fig.17. Export Button asked for Confirm & Save

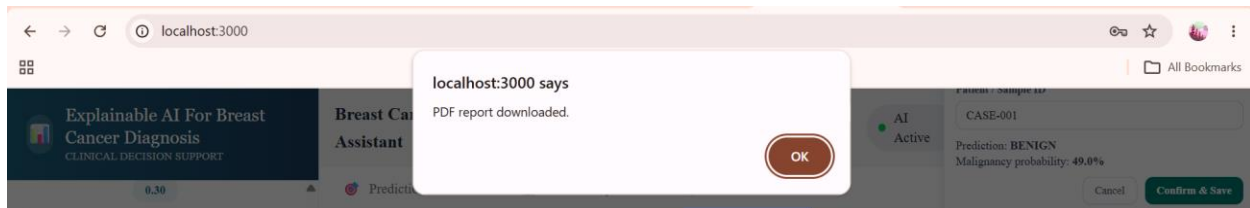


Fig.18. Pdf downloaded when clicked on save

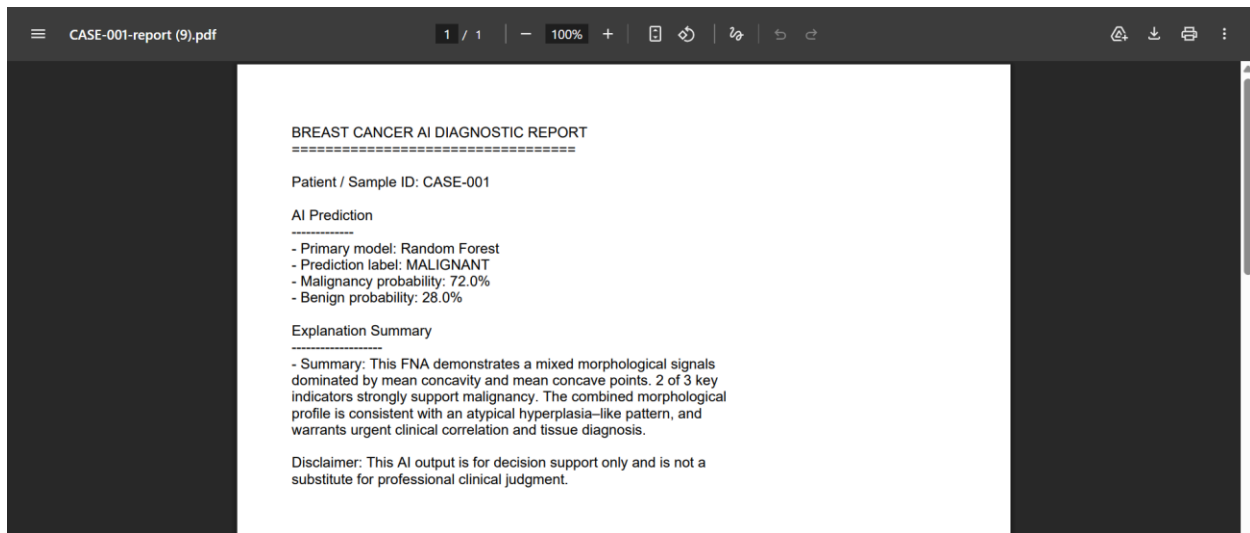


Fig.19. Diagnosis Report

Access to the dashboard is restricted to authenticated users. A lightweight signup and login flow backed by MongoDB Atlas stores user credentials, and the Next.js frontend enforces guarded routes that block unauthorised access to prediction, explanation, model-confidence and report-export pages, illustrating how role-based or identity-provider-based schemes could be integrated in future work.

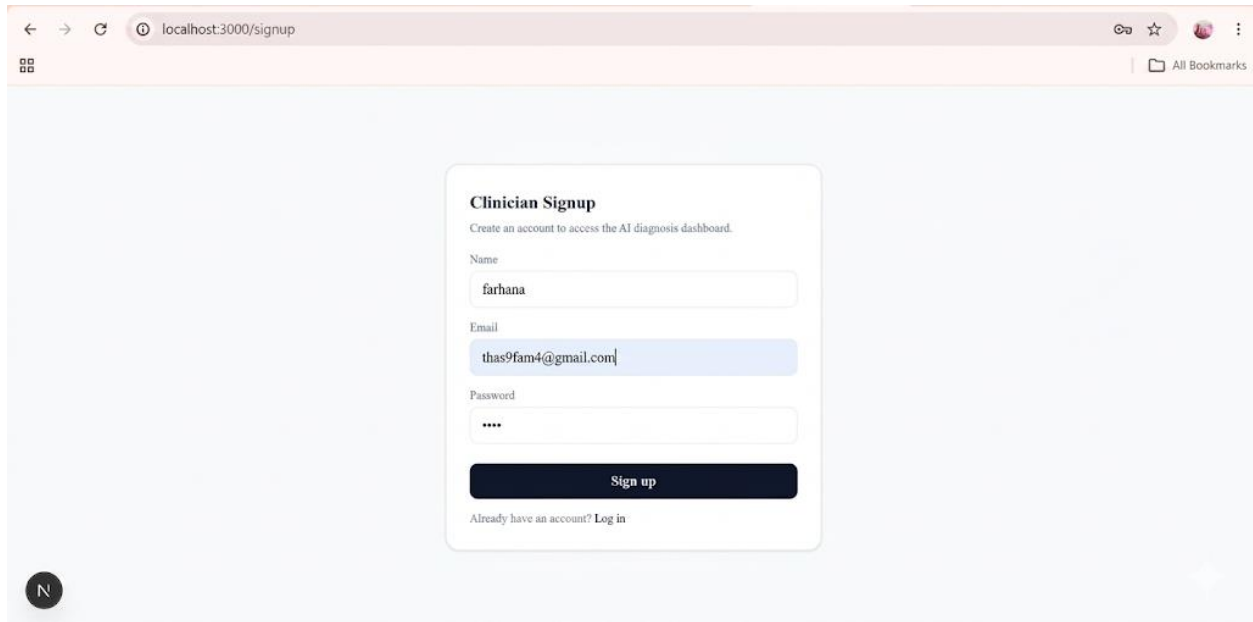


Fig.20. Signup

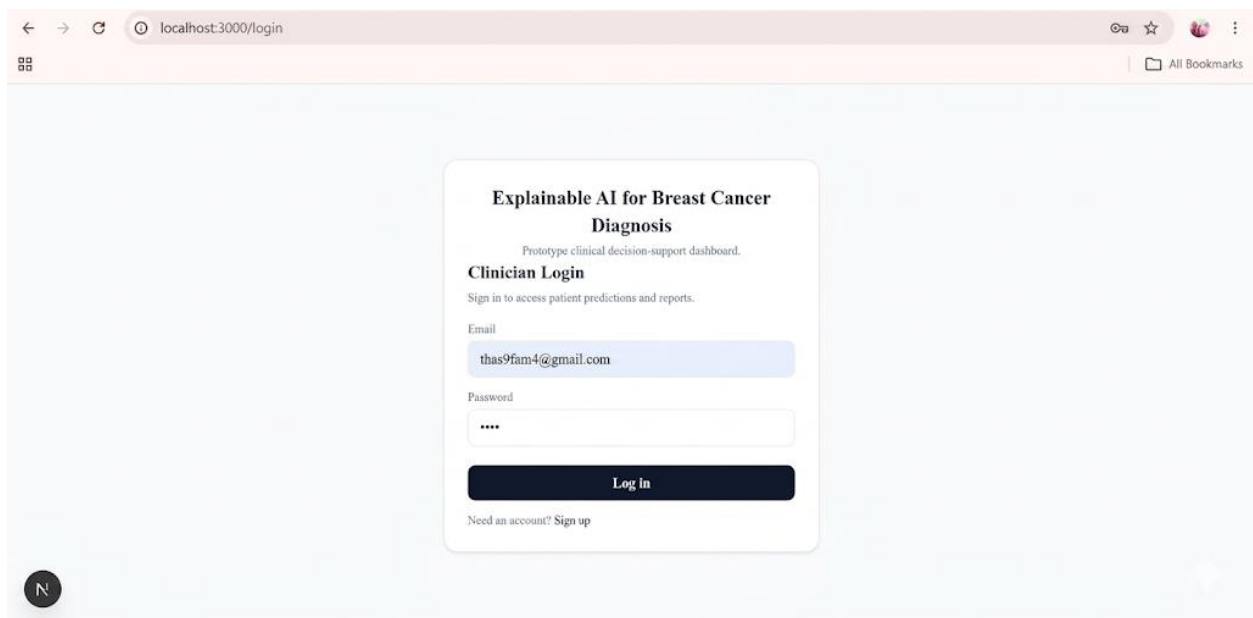


Fig.21.Login

4.4 Database, security and deployment considerations

The prototype uses MongoDB Atlas as a managed cloud database to persist user authentication data (signup and login credentials) and basic configuration. Although the dataset used for model training is public and not stored in the database, Atlas is configured following standard security guidance: a project-specific database user with limited privileges is created; the cluster is restricted using an IP access list so that only the local development environment can connect;

and all client connections use TLS for encryption in transit (Atlas Data Encryption Guidance, 2024). Atlas also encrypts data at rest by default, providing additional protection against unauthorised access to stored information. These measures are included to simulate realistic security constraints, even though the prototype is only exercised in a controlled, non-clinical setting.

The current system is not deployed as a live, internet-accessible service. Instead, the FastAPI backend is run locally, optionally within a Docker container that bundles the Python runtime, dependencies and application code, while the Next.js frontend is served in development mode or as a locally hosted build. This arrangement allows the prototype to be demonstrated and evaluated on a single workstation without the complexities of cloud deployment, while still preserving a clear separation between frontend, backend and database layers (Prototypical Decision-Supporting System, 2020).

Performance considerations primarily concern the cost of SHAP value computation. By restricting explanations to one case at a time, using an efficient tree-based SHAP explainer where possible and limiting the size of the background dataset, the system achieves interactive response times suitable for exploratory use in this local environment (Explainable AI for Healthcare Applications, 2023; Explainable AI in Healthcare, 2025).

5. Results

5.1 Model performance

Three supervised learning models—Random Forest, XGBoost and Logistic Regression—were trained on the WDBC training set and evaluated on the held-out test set using the metrics defined in Chapter 3. This comparative approach follows many studies that benchmark multiple algorithms on breast-cancer datasets rather than committing to a single model (Predicting Breast Cancer Leveraging Supervised ML, 2022; Feature Selection Using Genetic Algorithm, 2016). Table.2. summarises the performance of the three models in terms of accuracy, sensitivity, specificity, AUC-ROC, Brier score and false negative rate (FNR).

XGBoost achieved the highest overall accuracy (97.37%) and the lowest Brier score (0.0275), indicating both strong discrimination and good probabilistic calibration (The Application and Comparison of Machine Learning Models, 2022). It also attained the highest sensitivity (98.61%) and a relatively high specificity (95.24%), resulting in the lowest false negative rate (1.39%). Random Forest performed slightly worse but still competitively, while Logistic Regression achieved the lowest accuracy and highest FNR among the three, although its AUC-ROC remained high at 0.993, reflecting good ranking ability. These results are broadly consistent with prior work that reports strong performance for ensemble methods on the WDBC dataset (Predicting Breast Cancer Leveraging Supervised ML, 2022; Feature Selection Using Genetic Algorithm, 2016).

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)	AUC-ROC	Brier score	False Negative Rate(%)
Random Forest	94.74	95.83	92.86	0.992	0.0324	4.17
XG Boost	97.37	98.61	95.24	0.994	0.0275	1.39
Logistic Regression	93.86	93.06	95.24	0.993	0.0385	6.94

Table.2. Performance of candidate models on the WDBC test set (accuracy, sensitivity, brier score, specificity, FNR, AUC-ROC).

The metrics in Table 2 are consistent with the results obtained during the initial Colab experiments, where the same three models were tuned and evaluated using train/test accuracy and confusion matrices before export to the backend.

5.2 Confusion matrices and error patterns

To better understand the clinical implications of these results, confusion matrices were computed for each model. The key quantity of interest is the number and proportion of false negatives, since missed malignant cases are more serious than false positives in a diagnostic context (Predicting Breast Cancer Leveraging Supervised ML, 2022). The false negative rate values reported in Table 2 show that XGBoost produces the fewest false negatives (1.39% of malignant cases), followed by Random Forest (4.17%), while Logistic Regression has the highest false negative rate (6.94%).

These differences suggest that ensemble models, particularly XGBoost, may offer a more favourable trade-off between sensitivity and specificity on this dataset, although all three models achieve high AUC-ROC and good Brier scores. In line with design-science recommendations, the project does not fix a single model as “final” but instead exposes performance metrics, confusion matrices and explanations for all three, allowing future stakeholders to decide which operating point and algorithm best align with their risk tolerance and workflow (Design Science Methodology, 2025). Within the prototype, these confusion matrices and metrics are aggregated in the **Model confidence** tab, where they are combined with per-case agreement information indicating whether the models concur or disagree on individual predictions.

5.3 Global feature importance with SHAP

Global SHAP analysis is performed by computing SHAP values for all instances in the test set and aggregating them to derive feature-importance scores. This analysis highlights which WDBC features contribute most strongly, on average, to the models’ predictions of malignancy versus benignity (Explainable AI in Healthcare, 2025; Explainable AI for Healthcare Applications, 2023). In line with previous work on feature selection and importance in this dataset, features related

to cell size, shape irregularity and concavity—such as worst radius, worst perimeter and worst concave points—tend to appear among the most influential variables (Feature Selection Using Genetic Algorithm, 2016; GeeksforGeeks, 2024).

In this project, global SHAP analyses are computed for **each of the three models**, and the resulting feature-importance rankings are presented in the dedicated **Feature importance** tab of the dashboard. For each model, the tab shows how the ten WDBC feature groups contribute to predictions across the test set, allowing users to compare, for example, whether XGBoost relies more heavily on worst concave points than Random Forest or Logistic Regression. The SHAP summary plots reveal that higher values of the key features generally push predictions towards the malignant class, while lower values push them towards benign, which is consistent with cytological expectations that larger and more irregular nuclei are characteristic of malignancy (UCI Machine Learning Repository, 2021; GeeksforGeeks, 2024). Other features, such as smoothness or fractal dimension, may play a secondary but still meaningful role, contributing to fine-grained risk adjustments in borderline cases (Feature Selection Using Genetic Algorithm, 2016).

5.4 Local explanations and case-level insights

Local SHAP explanations are generated for individual test instances and for example cases submitted through the prototype interface. For each case, a SHAP force plot or equivalent visualisation shows how the baseline risk is adjusted by contributions from specific features, highlighting the main drivers of the predicted class (Explainable AI in Healthcare, 2025). For instance, a case predicted as malignant with high confidence may show strong positive contributions from features such as worst radius, worst area and worst concave points, while a benign case may be characterised by low values of the same features exerting negative contributions towards malignancy (Feature Selection Using Genetic Algorithm, 2016; WDBC GFS Study, 2021).

These local explanations also help to identify borderline cases where the overall prediction probability is closer to 0.5 and feature contributions partially offset one another. In such situations, the explanation can reveal, for example, that high values of one suspicious feature are counterbalanced by benign-leaning values of others, which might prompt more cautious clinical interpretation (Explainable AI in Healthcare, 2025). This dual role of SHAP—highlighting both clear-cut and ambiguous patterns—is echoed in recent reviews of XAI for breast-cancer detection and risk assessment (Explainable AI in Breast Cancer Detection and Risk, 2024). Within the prototype, these local explanations are coupled with the **Export report** feature: when a user exports a case report, the diagnostic PDF includes not only the primary model's label and probabilities but also a concise narrative summary distilled from the local SHAP contributions, as illustrated in the sample report for CASE-001.

5.5 Prototype behaviour in use

From a system perspective, the prototype successfully integrates model predictions, SHAP explanations and performance summaries into a coherent workflow. During testing, prediction and explanation responses for single cases are generated within an acceptable time frame for interactive use, with end-to-end latency suitable for on-demand decision support on a standard workstation. This performance is consistent with other SHAP-based decision-support tools reported in the literature, which also emphasise single-case, on-demand explanations (Breast Cancer Prediction SHAP Dashboard, 2025; Explainable AI in Healthcare, 2025).

The explanation views allow users to switch between a global overview of feature importance, local case-specific breakdowns and aggregated model-level information in the **Model confidence** tab. Informal testing suggests that this combination provides a more intuitive understanding of model behaviour than numeric metrics alone, particularly when accompanied by the narrative summaries embedded in the exported diagnostic reports (Breast Cancer Prediction SHAP Dashboard, 2025; Supporting Breast Cancer Decisions, 2017). In a typical usage scenario, a user inputs features, inspects the prediction and local SHAP explanation, checks the Model confidence tab to understand how all three models behave on the test set and in terms of ensemble agreement, and finally uses the Export report function to generate a structured case report for documentation or discussion. A more formal usability evaluation with clinicians would be needed to confirm these observations.

Throughout these interactions, the availability of multiple explanation modes on the prediction dashboard, the FNA-style clinical pattern-match summaries, the sidebar with clinical reference ranges and the model-toggle and filtering options in the top navigation collectively demonstrate concrete progress towards the project's aims of integrating explainable models into a clinician-oriented interface, rather than remaining at the level of abstract metrics or offline notebooks.

The current interface design is, however, only a first step towards a robust clinician-facing tool. It has not been formally evaluated with end-users, and important aspects such as integration with existing workflow systems, logging of decisions and support for multi-disciplinary team discussions remain unexplored. XAI-focused evaluations also highlight that interpretations can be misunderstood if users are not adequately trained in how to read explanation plots, which suggests that future iterations should include inline guidance and possibly alternative explanation formats tailored to different user roles (Explainable AI in Healthcare, 2025).

6. Discussion

6.1 Interpretation of predictive performance

The predictive performance achieved in this project is broadly comparable to that reported in recent studies that apply supervised learning to the WDBC dataset. Accuracies in the mid- to high-90% range and AUC values close to 1.0 are consistent with the performance of algorithms such as support vector machines, random forests and gradient boosting reported in the

literature (Predicting Breast Cancer Leveraging Supervised ML, 2022; Feature Selection Using Genetic Algorithm, 2016). This alignment provides some reassurance that the implementation and evaluation procedures used here are reasonable and that the three evaluated models are competitive on this benchmark.

From a clinical perspective, however, raw accuracy is less informative than the balance between sensitivity and specificity. Given the high cost of false negatives, it is important that models maintain very high sensitivity, even if this is accompanied by a modest increase in false positives. The confusion-matrix analysis indicates that XGBoost offers the lowest false negative rate, with Random Forest and Logistic Regression performing slightly worse but still achieving high sensitivity values. By presenting these trade-offs explicitly in the **Model confidence** tab, the prototype encourages users to consider how different models balance missed malignancies against unnecessary work-ups, rather than treating the output of a single algorithm as definitive (Supporting Breast Cancer Decisions, 2017; Prototypical Decision-Supporting System, 2020).

6.2 Interpretation of SHAP-based explanations

The SHAP analyses provide insight into whether the models behave in ways that are consistent with established cytological understanding. The prominence of features related to cell size and shape irregularity—such as worst radius, worst perimeter and worst concave points—in global SHAP rankings aligns with previous feature-selection studies on the WDBC dataset, which also identify these variables as highly informative for malignancy classification (Feature Selection Using Genetic Algorithm, 2016; WDBC GFS Study, 2021). This concordance suggests that the models are capturing clinically plausible patterns rather than relying purely on artefacts or noise.

At the local level, SHAP explanations help to make individual predictions more transparent by showing how specific feature values influenced the outcome. For straightforward cases, such as clearly malignant or clearly benign examples, the explanations reinforce the prediction by highlighting consistent, strongly weighted features. For borderline cases, they reveal competing influences that may prompt more cautious interpretation and potentially further investigation. This ability to expose uncertainty and conflict in the feature space is one of the reasons SHAP is recommended in XAI guidelines for healthcare applications (Explainable AI in Healthcare, 2025; Explainable AI in Breast Cancer Detection and Risk, 2024). In the prototype, these local SHAP insights are distilled into short narrative summaries that appear both in the prediction panel and in the exported diagnostic reports, illustrating how technical explanations can be translated into domain-appropriate language.

6.3 Evaluation of the decision-support prototype

Beyond numerical performance and explanation fidelity, the usefulness of the system depends on how well the interface communicates information to its intended users. The prototype provides a simple, task-oriented workflow: enter or review features, obtain a prediction,

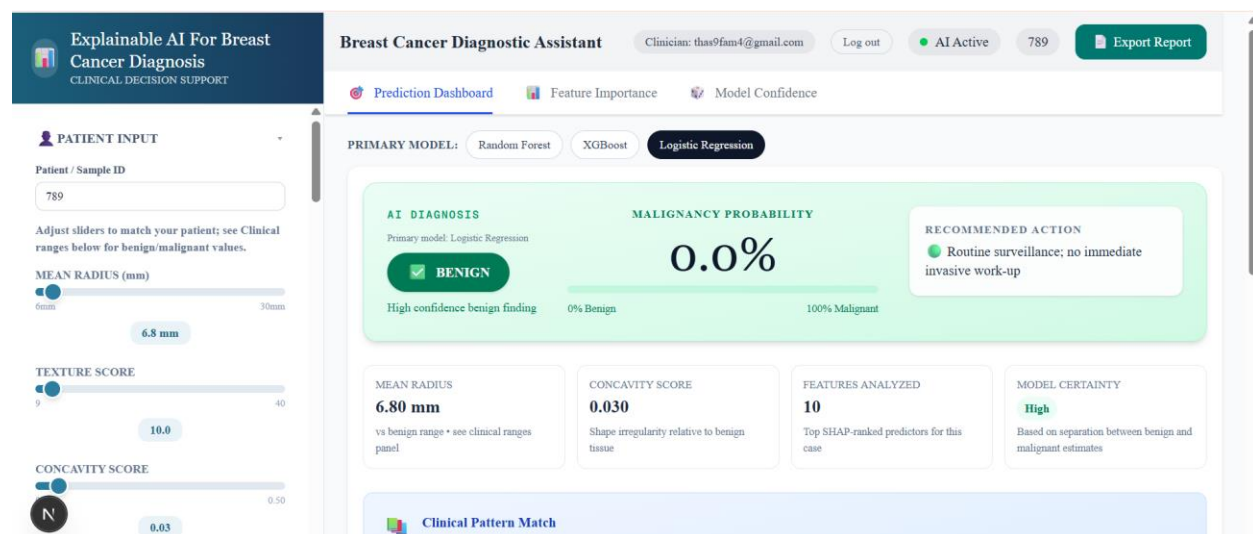
inspect local SHAP explanations, explore global feature importance across all three models in the **Feature importance** tab and review model-level behaviour in the **Model confidence** tab. This is broadly consistent with recommendations from studies of breast-cancer decision-support systems, which emphasise transparent presentation of key data and options rather than complex algorithmic details (Supporting Breast Cancer Decisions, 2017; Prototypical Decision-Supporting System, 2020).

The inclusion of an **Export report** function further supports potential integration into clinical workflows by packaging the most important outputs—primary model, predicted label and probabilities and a concise explanation summary—into a structured artefact suitable for documentation or case discussion, as demonstrated by the CASE-001 report. However, the current interface design is only a first step towards a robust clinician-facing tool. It has not been formally evaluated with end-users, and important aspects such as integration with existing workflow systems, logging of decisions and support for multi-disciplinary team discussions remain unexplored. XAI-focused evaluations also highlight that interpretations can be misunderstood if users are not adequately trained in how to read explanation plots, which suggests that future iterations should include inline guidance and possibly alternative explanation formats tailored to different user roles (Explainable AI in Healthcare, 2025).

Test Case Scenarios

To make the evaluation more concrete, several representative test scenarios were exercised using the running prototype:

- A clearly benign case with low values for radius, concavity and concave points produced a BENIGN prediction with low malignancy probability, SHAP attributions dominated by features pushing towards benignity and a narrative summary that aligned with benign FNA patterns. (Using logistic regression)



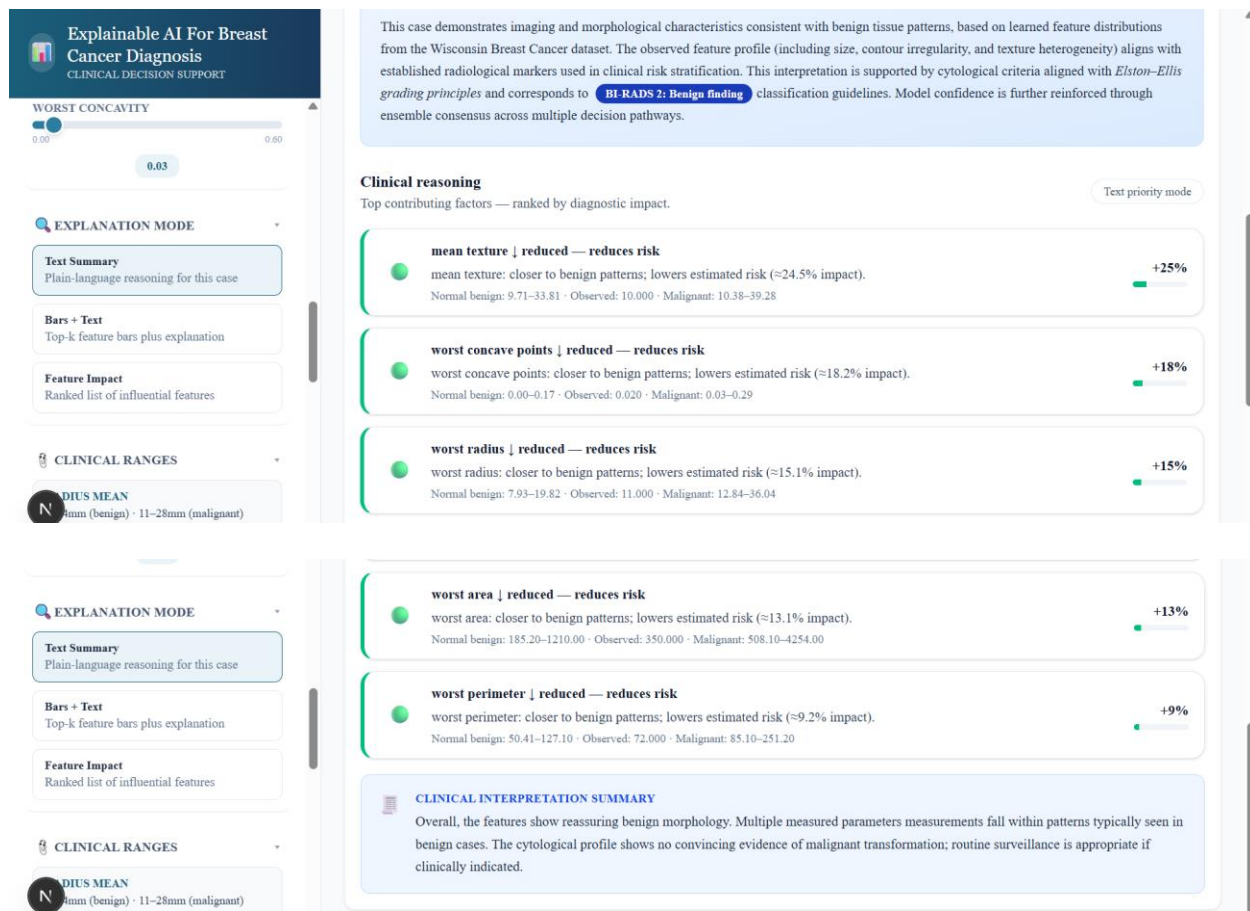


Fig.22. Explanation Mode 1 (“Clinical reasoning – text summary”)

- A clearly malignant case with markedly elevated worst radius, worst perimeter and worst concave points was classified as MALIGNANT with high malignancy probability; SHAP explanations highlighted these features in red as high-risk drivers, and the clinical summary emphasised an aggressive pattern consistent with malignant cytology.
- A borderline case with intermediate feature values yielded an intermediate malignancy probability, mixed SHAP contributions (some towards benign, some towards malignant) and explanatory text that explicitly noted diagnostic uncertainty and recommended closer clinical correlation.

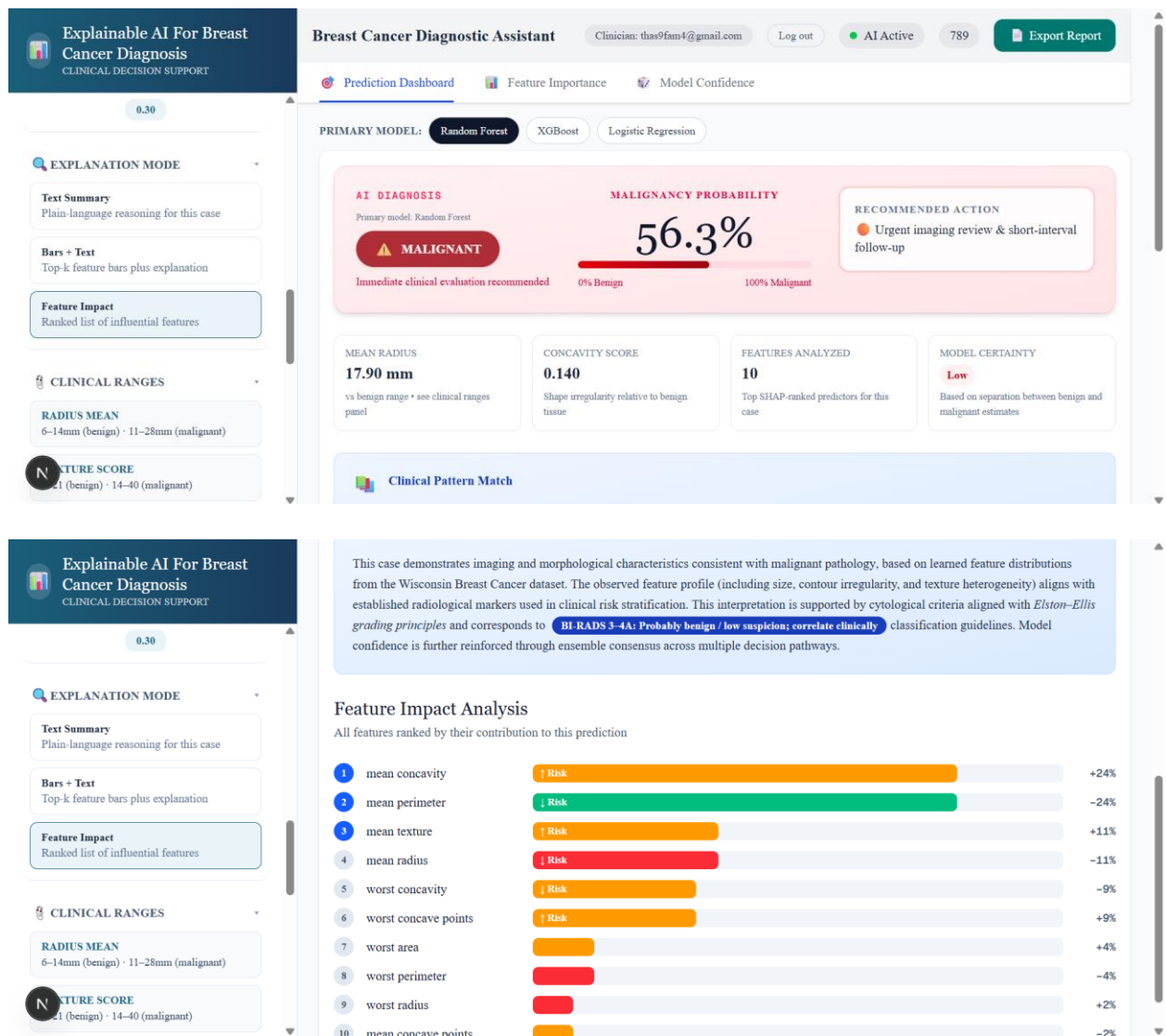


Fig.23. Explanation Mode 3 (“Feature impact analysis”)- shows low suspicion

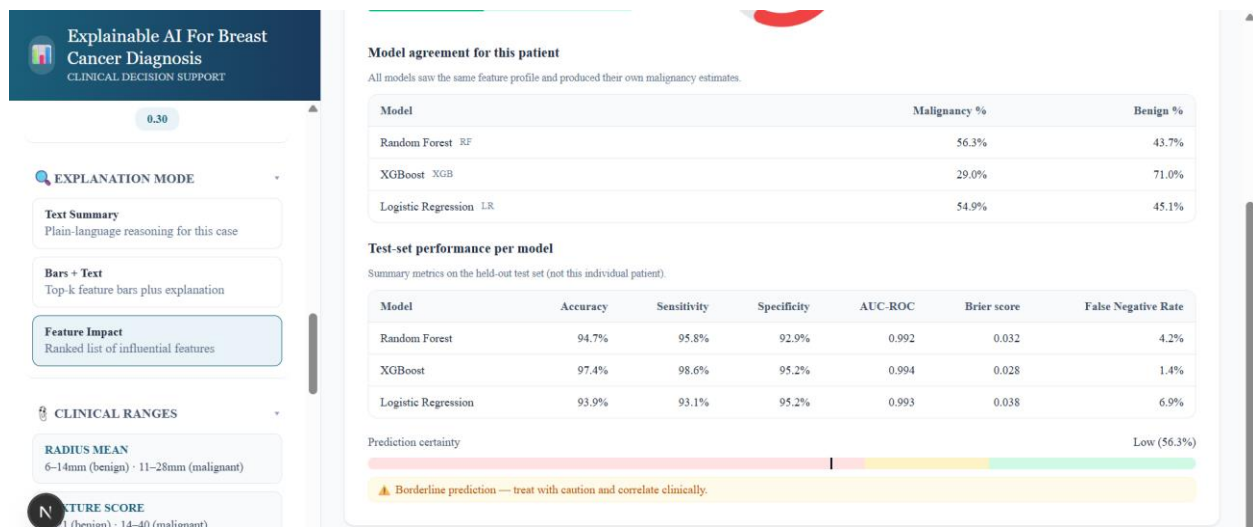


Fig.24. Model confidence tab-shows borderline prediction.

- Error-handling was checked by submitting incomplete or invalid payloads to the FastAPI endpoint; in these cases the backend returned informative HTTP error messages, which the frontend surfaced as clear user-facing notifications rather than failing silently.

6.4 Limitations

Several limitations of this work must be acknowledged. First, the study is based entirely on a single benchmark dataset drawn from a specific institution and time period, which raises questions about the external validity of both the models and their explanations when applied to other patient populations or imaging protocols (UCI Machine Learning Repository, 2021; Predicting Breast Cancer Leveraging Supervised ML, 2022). Second, the dataset focuses on FNA-derived features and does not include additional clinical variables or imaging findings that would normally inform real-world decisions, so the models capture only a subset of the relevant information (UCI Machine Learning Repository, 2021).

Third, the evaluation is primarily technical, relying on offline test-set metrics and qualitative inspection of explanations via the dashboard and exported reports. While this is sufficient for a design-science prototype, it falls short of the human-centred evaluations needed before an XAI-enabled decision-support system can be considered for clinical use (Explainable AI in Healthcare, 2025; Supporting Breast Cancer Decisions, 2017). Finally, the interface and explanation designs are tailored to the WDBC feature set and to the current trio of models; they would need to be revisited if the system were extended to handle raw images, additional clinical data or a broader set of algorithms, as suggested by current trends in breast-cancer XAI research (Explainable AI for Healthcare Applications, 2023; Explainable AI in Breast Cancer Detection and Risk, 2024).

[Table 3: Summary of key limitations grouped by data, models, explanations, interface and evaluation dimensions (placeholder).]

7. Conclusions and Future Work

7.1 Conclusions

This project set out to develop and evaluate an explainable machine-learning–based decision-support prototype for breast cancer diagnosis using FNA-derived features from the WDBC dataset. Rather than committing to a single best model, three supervised classifiers—Random Forest, XGBoost and Logistic Regression—were trained and compared on a held-out test set using accuracy, sensitivity, specificity, AUC-ROC, Brier score and false negative rate as evaluation metrics (Predicting Breast Cancer Leveraging Supervised ML, 2022; Feature Selection Using Genetic Algorithm, 2016).

The comparative results showed that all three models achieved high discrimination, with AUC-ROC values above 0.99, and good calibration as indicated by low Brier scores. XGBoost provided the strongest overall performance, with the highest accuracy (97.37%), highest sensitivity (98.61%) and lowest false negative rate (1.39%), while Random Forest and Logistic Regression performed slightly less well but still reached levels comparable to prior WDBC studies. This confirms that the modelling and evaluation pipeline implemented in this project behaves as expected for this benchmark problem. SHAP analyses confirmed that the model’s behaviour is consistent with cytological expectations, highlighting features related to cell size and shape irregularity as primary drivers of malignancy predictions.

From an engineering perspective, the project demonstrates that a pragmatic full-stack architecture based on a FastAPI backend (Python, SHAP, LIME, scikit-learn, TensorFlow-CPU, xgboost and joblib) and a Next.js frontend (TypeScript and CSS) can support interactive, SHAP-enhanced explanations in a local development environment. The use of MongoDB Atlas with IP allow-listing, least-privilege database users, TLS and encryption at rest illustrates how realistic security configurations can be incorporated even in a prototype that is not deployed as a live internet-facing service (Atlas Data Encryption Guidance, 2024). Overall, the system provides a concrete instantiation of design-science principles for explainable breast-cancer decision support without crossing into production deployment.

[Figure 18: Conceptual summary of the project’s contributions – model, SHAP explanations, Next.js/FastAPI prototype and secure MongoDB Atlas backend (placeholder).]

7.2 Contributions

Beyond the conceptual framing, the project makes several tangible contributions:

- **End-to-end XAI pipeline:** A complete, documented pipeline from WDBC data ingestion to a FastAPI service that exposes prediction and SHAP explanation endpoints for all three evaluated models, integrated with a Next.js dashboard running in a local development environment.
- **SHAP integration pattern for ensemble models:** A reusable implementation of SHAP for tree-based models (including xgboost) using a compact background dataset, matching the dominant approach in recent breast-cancer XAI work.
- **Clinician-oriented Next.js dashboard:** A TypeScript-based dashboard that separates global performance metrics, global feature-importance views and local, patient-specific explanations, inspired by recent SHAP dashboards for breast-cancer risk prediction.
- **Security-aware data layer using MongoDB Atlas:** A configuration that applies IP-based access control, TLS for encryption in transit and encryption at rest by default, following Atlas guidance, and uses a dedicated database user with limited privileges.
- **Containerized backend services:** A Dockerised FastAPI backend with explicit dependency management, demonstrating how XAI-enabled ML services can be packaged for local or cloud-based testing and forming a basis for more advanced deployment pipelines.
- **A set of dedicated dashboard tabs** that go beyond a simple prediction view: a Feature importance tab that surfaces global SHAP feature-importance plots for the ten WDBC feature groups across all three evaluated models; a Model confidence tab that summarises test-set performance, confusion matrices and ensemble agreement; and an Export report function that generates structured diagnostic reports containing the primary model, probabilities and narrative explanation for individual cases.

These contributions collectively move from abstract discussion of explainable AI for breast-cancer diagnosis towards a concrete software artefact that can be inspected, extended and evaluated by other researchers and practitioners.

7.3 Future Work

7.3.1 Enhancing explanations and the user interface

The current dashboard focuses on SHAP explanations and generated summaries. Future improvements could include multiple explanation methods to improve interpretability and user trust:

- Adding *what-if analysis* tools in the Next.js interface to allow clinicians to modify WDBC features and observe changes in predictions and SHAP values.
- Implementing *similar case retrieval* using de-identified historical cases stored in MongoDB to support example-based reasoning.
- Creating *role-specific interfaces*, providing simplified views for general users and detailed SHAP visualisations for specialists.

7.3.2 Data and model extensions

Several improvements could strengthen the model and data framework:

- Integrating additional clinical and imaging features to improve prediction performance.
- Introducing model versioning and a registry system to compare multiple deployed models.
- Developing automated retraining pipelines for model updates and performance monitoring.

7.3.3 Security, deployment and validation

Further development towards real-world deployment would require stronger security and validation measures:

- Expanding authentication with role-based access control and institutional identity systems.
- Extending deployment into development, staging and production environments with automated testing.
- Conducting formal clinician user studies to evaluate explanation clarity, usefulness and trust.

Addressing these areas could support the transition from a functional prototype to a clinically evaluated explainable decision-support system.

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Appendix

Appendix A: Weekly Progress Form

Weekly Progress Logbook - Semester 2 (Feb 2026 to May 2026)			
Student ID:	32146925		
Student Name:	Thaspeeha Vahithu		
Course:	Bsc(Hons) Computer Science		
Project Title:	Explainable AI For Breast Cancer Diagnosis		
Supervisor Name:	Mr. Mohammed Rafiq Firdouz		
w/c (Mon)	Week	Weekly log of progress	Supervisor Initials
2/9/2026	1	Dataset understanding and loading	h.v.
2/16/2026	2	Feature selection and baseline models	h.v.
2/23/2026	3	Full evaluation metrics	h.v.
3/2/2026	3	Model comparison	h.v.
3/9/2026	4	Tools like SHAP used for explainability (local and global)	h.v.
3/16/2026	5	Backend API (FastAPI) and data layer (MongoDB)	h.v.
3/23/2026	6	Frontend dashboard (Next.js) & local explanations	h.v.
Spring break			
4/6/2026	7	Adding Feature: Global importance and model confidence	h.v.
4/13/2026	9	Security & polishing	h.v.
4/20/2026	10	Documentation-1	h.v.
4/27/2026	11	Documentation-2	h.v.
5/1/2026	12	Documentation-3	h.v.
5/15/2026	13	Presentation	h.v.